

# Tree water deficit: data exploration

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```
require(dplyr)
require(ggplot2)
require(car)
require(stringr)
require(lubridate)
require(zoo)
#require(flextable)
library(broom)
library(ggplot2)
library(forecast)

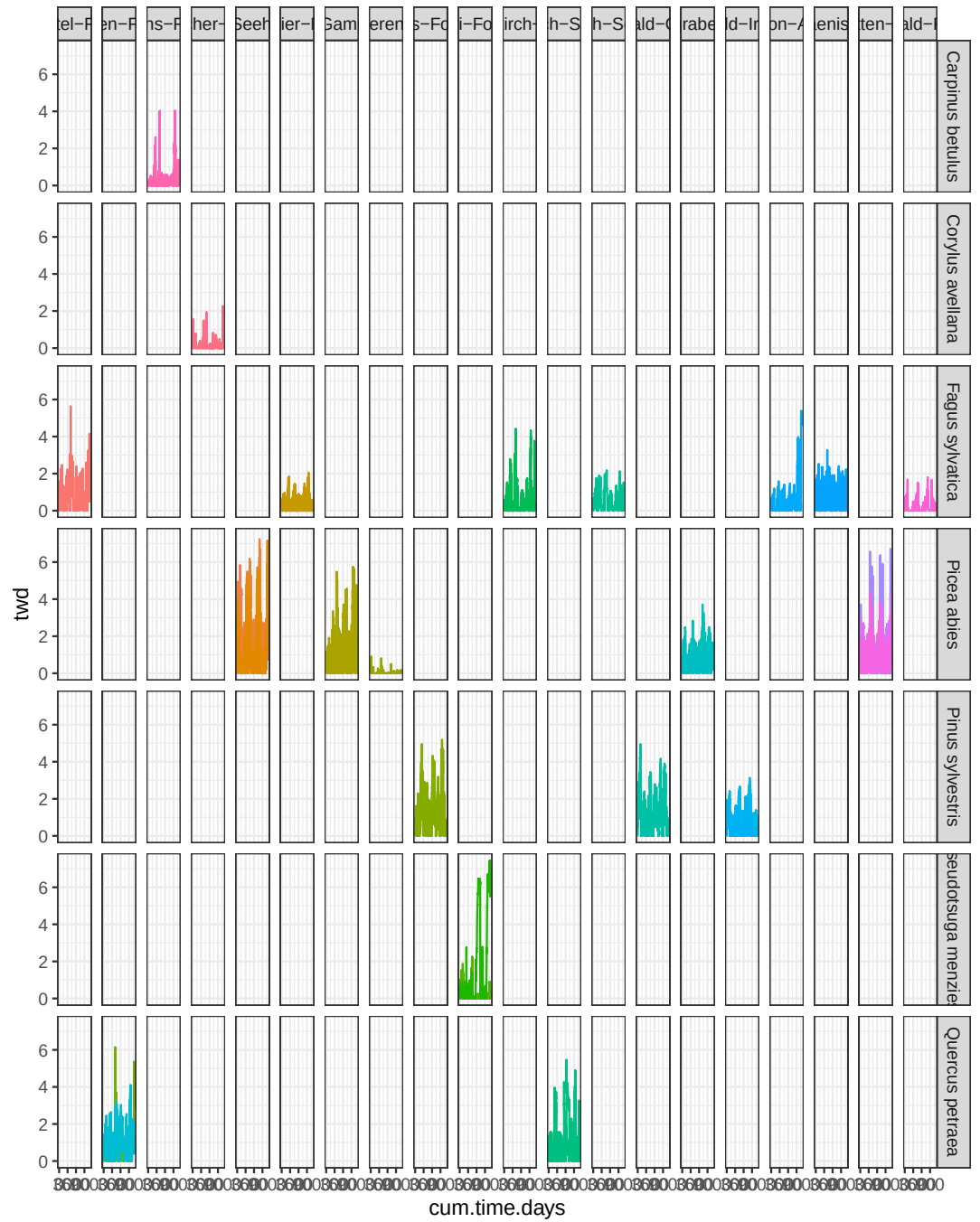
knitr::opts_chunk$set(fig.width=7, fig.height=7)
```

```
bool_run_dataset <- TRUE

if(bool_run_dataset){source("read.datasets.r")}
else {
  df.ts.series2 <- read.csv("df.ts.series2.csv")
  num.cols <- c("twd", "pr", "at", "ws", "dp", "sr", "lr", "day.of.year")
}
```

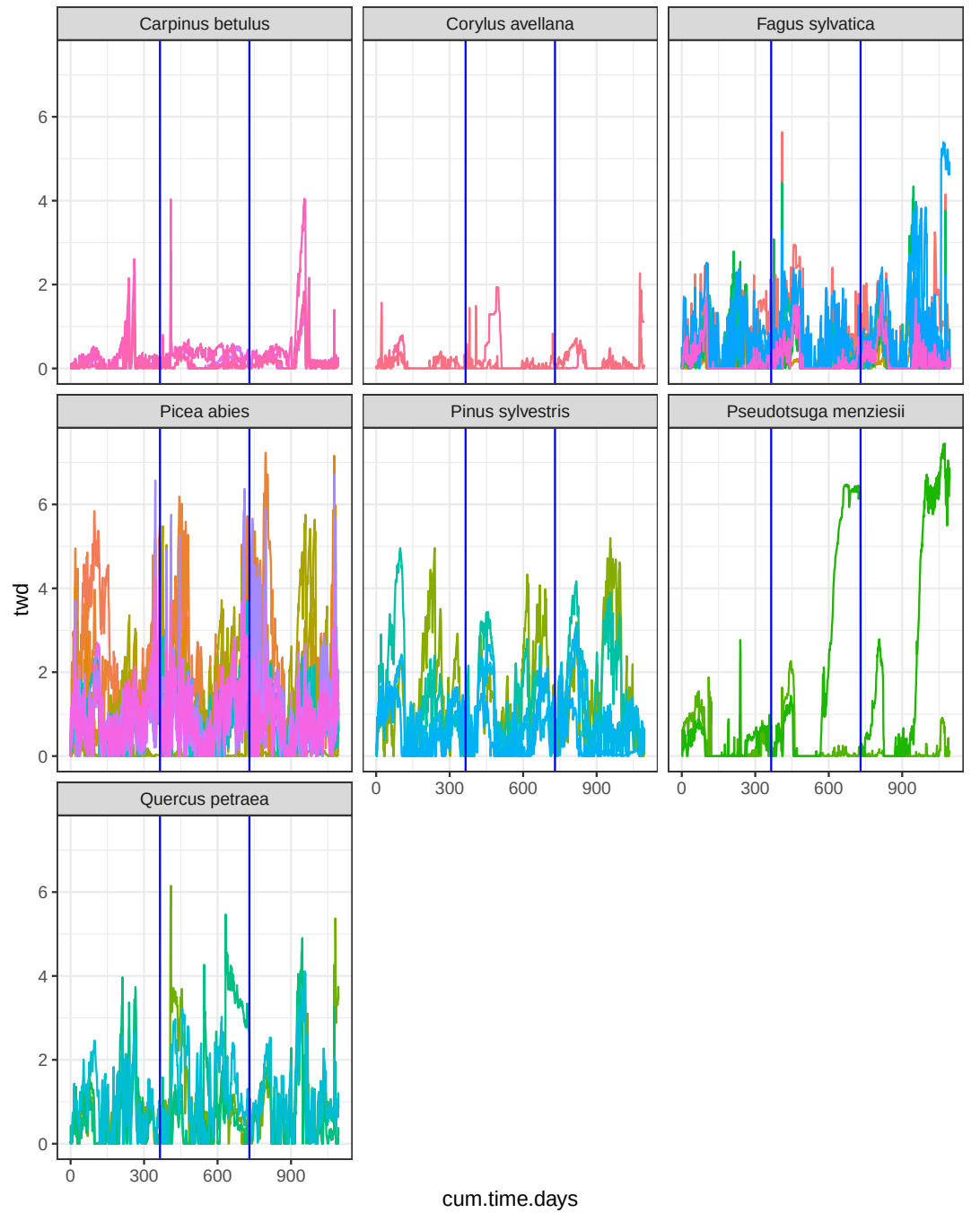
```
plot_basic <- ggplot(df.ts.series2, aes(x = cum.time.days, y = twd, group = tree.id)) +
  geom_line(aes(color = tree.id)) +
  theme_bw() +
  theme(legend.position='bottom') +
  guides(color = guide_legend(nrow = 3))

plot_basic +
  facet_grid(vars(species), vars(site))
```



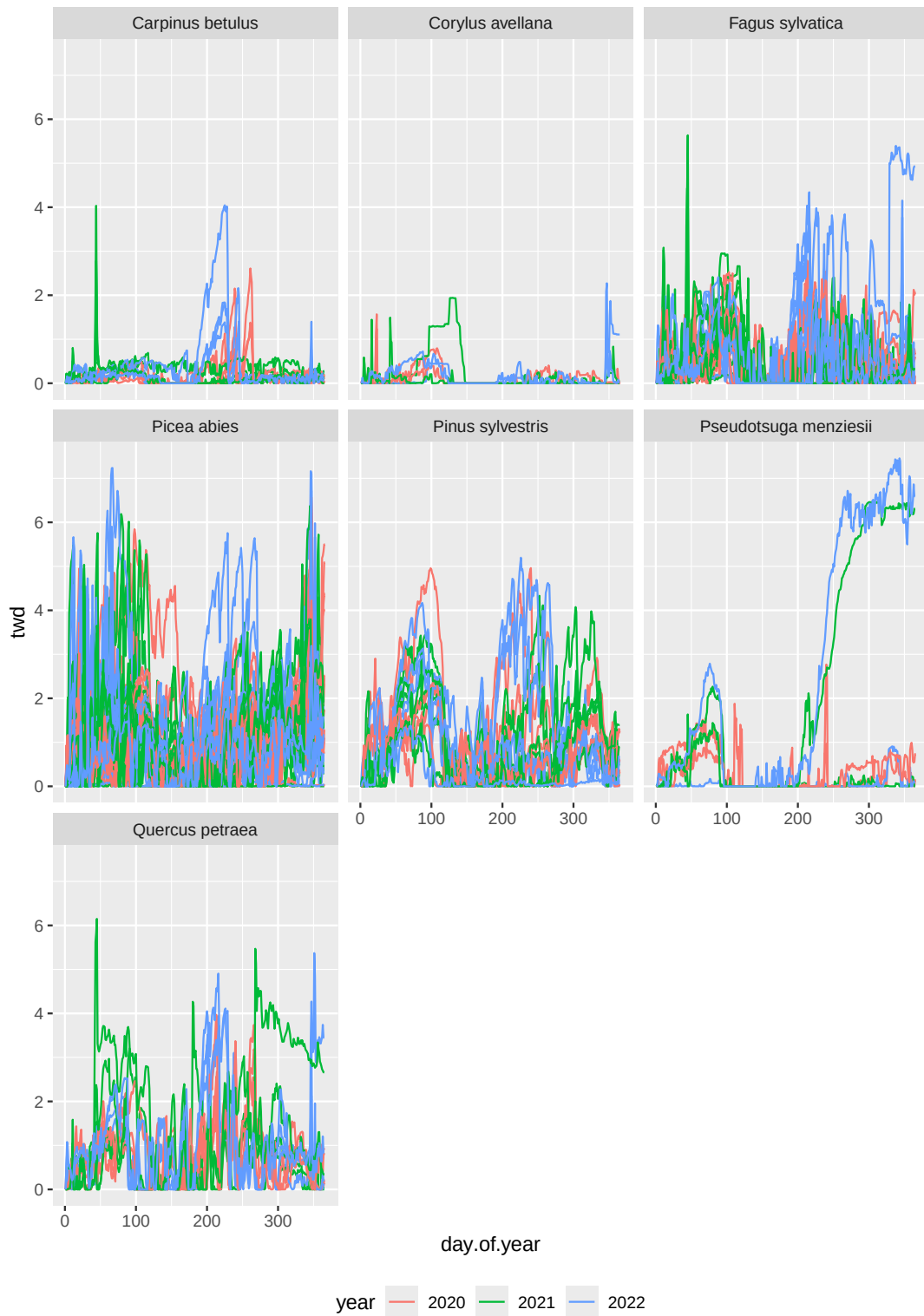
1 12 15 18 20 23 26 29 31 34 37 4 42  
 d 10 13 16 19 21 24 27 3 32 35 38 40 5  
 11 14 17 2 22 25 28 30 33 36 39 41 6

```
plot_basic +  
  geom_vline(xintercept = 365, color = "blue") +  
  geom_vline(xintercept = 730, color = "blue") +  
  facet_wrap(~species)
```





```
ggplot(  
  df.ts.series2,  
  aes(x = day.of.year, y = twd,  
    group = interaction(year, tree.id))) +  
  geom_line(aes(color = year)) +  
  facet_wrap(~species) +  
  theme(legend.position='bottom')
```



```
#lapply(df.ts.series2, class)

num.vars.df <- df.ts.series2[,num.cols]
cor_matrix <- cor(num.vars.df, use = "complete.obs")
#flectable(cor_matrix)
cor_matrix
```

|             | twd          | pr           | at           | ws          | dp          |
|-------------|--------------|--------------|--------------|-------------|-------------|
| twd         | 1.000000000  | -0.131262923 | -0.119470916 | -0.12483220 | -0.22037315 |
| pr          | -0.131262923 | 1.000000000  | 0.002590598  | 0.08997804  | 0.16716389  |
| at          | -0.119470916 | 0.002590598  | 1.000000000  | 0.08083005  | 0.90584762  |
| ws          | -0.124832202 | 0.089978045  | 0.080830051  | 1.00000000  | 0.02792756  |
| dp          | -0.220373147 | 0.167163893  | 0.905847623  | 0.02792756  | 1.00000000  |
| sr          | 0.008947053  | -0.276992037 | 0.614793728  | 0.04611393  | 0.38822556  |
| lr          | -0.198775942 | 0.344976601  | 0.764139599  | 0.08014614  | 0.88312535  |
| day.of.year | 0.053890048  | 0.032564700  | 0.189554518  | -0.13950669 | 0.28942973  |

|             | sr           | lr          | day.of.year |
|-------------|--------------|-------------|-------------|
| twd         | 0.008947053  | -0.19877594 | 0.05389005  |
| pr          | -0.276992037 | 0.34497660  | 0.03256470  |
| at          | 0.614793728  | 0.76413960  | 0.18955452  |
| ws          | 0.046113925  | 0.08014614  | -0.13950669 |
| dp          | 0.388225562  | 0.88312535  | 0.28942973  |
| sr          | 1.000000000  | 0.16082884  | -0.17224888 |
| lr          | 0.160828837  | 1.00000000  | 0.26590182  |
| day.of.year | -0.172248882 | 0.26590182  | 1.00000000  |

```
#pairs(num.vars.df)
```

```
list_species <- unique(df.ts.series2$species)

for(iter_species in list_species){
  iter_df <- df.ts.series2 %>%
    filter(species == iter_species)

  print(iter_species)
  print(cor(iter_df[,num.cols]))
}
```

```
[1] "Fagus sylvatica"
```

|     | twd         | pr          | at         | ws          | dp         |
|-----|-------------|-------------|------------|-------------|------------|
| twd | 1.000000000 | -0.18137627 | 0.02541095 | 0.009627288 | -0.1405810 |

|             |              |             |             |              |            |
|-------------|--------------|-------------|-------------|--------------|------------|
| pr          | -0.181376267 | 1.000000000 | -0.02701793 | 0.094676772  | 0.1350574  |
| at          | 0.025410953  | -0.02701793 | 1.000000000 | -0.035522186 | 0.9010553  |
| ws          | 0.009627288  | 0.09467677  | -0.03552219 | 1.000000000  | -0.1330824 |
| dp          | -0.140581022 | 0.13505742  | 0.90105530  | -0.133082399 | 1.0000000  |
| sr          | 0.132248373  | -0.29568173 | 0.67872372  | 0.067925142  | 0.4252621  |
| lr          | -0.172377333 | 0.34107498  | 0.72575852  | -0.040340079 | 0.8641839  |
| day.of.year | -0.045990926 | 0.03258924  | 0.18032814  | -0.189827925 | 0.3046255  |

|             |             |             |             |
|-------------|-------------|-------------|-------------|
|             | sr          | lr          | day.of.year |
| twd         | 0.13224837  | -0.17237733 | -0.04599093 |
| pr          | -0.29568173 | 0.34107498  | 0.03258924  |
| at          | 0.67872372  | 0.72575852  | 0.18032814  |
| ws          | 0.06792514  | -0.04034008 | -0.18982793 |
| dp          | 0.42526205  | 0.86418388  | 0.30462553  |
| sr          | 1.00000000  | 0.15077338  | -0.16281669 |
| lr          | 0.15077338  | 1.00000000  | 0.28660976  |
| day.of.year | -0.16281669 | 0.28660976  | 1.00000000  |

[1] "Quercus petraea"

|             |             |             |             |              |             |
|-------------|-------------|-------------|-------------|--------------|-------------|
|             | twd         | pr          | at          | ws           | dp          |
| twd         | 1.00000000  | -0.19798010 | 0.14448897  | -0.046200810 | 0.03319407  |
| pr          | -0.19798010 | 1.00000000  | 0.03682531  | 0.196605657  | 0.17324305  |
| at          | 0.14448897  | 0.03682531  | 1.00000000  | -0.115668256 | 0.91545158  |
| ws          | -0.04620081 | 0.19660566  | -0.11566826 | 1.000000000  | -0.16557647 |
| dp          | 0.03319407  | 0.17324305  | 0.91545158  | -0.165576472 | 1.00000000  |
| sr          | 0.15232695  | -0.24692796 | 0.67703942  | -0.080532493 | 0.43101430  |
| lr          | -0.03945657 | 0.36513048  | 0.73197573  | 0.007775018  | 0.86106870  |
| day.of.year | 0.08598431  | 0.04182131  | 0.17383576  | -0.228046338 | 0.30951107  |

|             |             |              |             |
|-------------|-------------|--------------|-------------|
|             | sr          | lr           | day.of.year |
| twd         | 0.15232695  | -0.039456566 | 0.08598431  |
| pr          | -0.24692796 | 0.365130482  | 0.04182131  |
| at          | 0.67703942  | 0.731975733  | 0.17383576  |
| ws          | -0.08053249 | 0.007775018  | -0.22804634 |
| dp          | 0.43101430  | 0.861068700  | 0.30951107  |
| sr          | 1.00000000  | 0.134456029  | -0.16368994 |
| lr          | 0.13445603  | 1.000000000  | 0.29969086  |
| day.of.year | -0.16368994 | 0.299690856  | 1.00000000  |

[1] "Carpinus betulus"

|     |             |             |            |             |             |
|-----|-------------|-------------|------------|-------------|-------------|
|     | twd         | pr          | at         | ws          | dp          |
| twd | 1.00000000  | -0.14113324 | 0.2115865  | 0.06602011  | 0.08622268  |
| pr  | -0.14113324 | 1.00000000  | -0.0223143 | 0.01205931  | 0.16028541  |
| at  | 0.21158648  | -0.02231430 | 1.00000000 | -0.13926057 | 0.89930083  |
| ws  | 0.06602011  | 0.01205931  | -0.1392606 | 1.000000000 | -0.23337661 |
| dp  | 0.08622268  | 0.16028541  | 0.8993008  | -0.23337661 | 1.00000000  |
| sr  | 0.21104317  | -0.34360907 | 0.6914149  | 0.07728482  | 0.40995724  |

|             |            |            |           |             |            |
|-------------|------------|------------|-----------|-------------|------------|
| lr          | 0.05302899 | 0.36085692 | 0.7467854 | -0.18408455 | 0.86892949 |
| day.of.year | 0.01610075 | 0.06524682 | 0.1779694 | -0.14404783 | 0.32035788 |

|             |             |             |             |
|-------------|-------------|-------------|-------------|
|             | sr          | lr          | day.of.year |
| twd         | 0.21104317  | 0.05302899  | 0.01610075  |
| pr          | -0.34360907 | 0.36085692  | 0.06524682  |
| at          | 0.69141491  | 0.74678538  | 0.17796937  |
| ws          | 0.07728482  | -0.18408455 | -0.14404783 |
| dp          | 0.40995724  | 0.86892949  | 0.32035788  |
| sr          | 1.00000000  | 0.17154998  | -0.16269968 |
| lr          | 0.17154998  | 1.00000000  | 0.28413664  |
| day.of.year | -0.16269968 | 0.28413664  | 1.00000000  |

[1] "Corylus avellana"

|             |             |             |             |             |            |
|-------------|-------------|-------------|-------------|-------------|------------|
|             | twd         | pr          | at          | ws          | dp         |
| twd         | 1.00000000  | -0.07676496 | -0.15355564 | 0.06325104  | -0.2518433 |
| pr          | -0.07676496 | 1.00000000  | 0.03552847  | 0.15382165  | 0.1728802  |
| at          | -0.15355564 | 0.03552847  | 1.00000000  | -0.22953204 | 0.9248918  |
| ws          | 0.06325104  | 0.15382165  | -0.22953204 | 1.00000000  | -0.2471297 |
| dp          | -0.25184328 | 0.17288019  | 0.92489175  | -0.24712970 | 1.0000000  |
| sr          | 0.07053484  | -0.26731509 | 0.68725381  | -0.15772881 | 0.4290405  |
| lr          | -0.23748086 | 0.38456779  | 0.69517075  | -0.10127215 | 0.8497979  |
| day.of.year | -0.14072546 | 0.03335630  | 0.17507330  | -0.21332874 | 0.3075513  |

|             |             |             |             |
|-------------|-------------|-------------|-------------|
|             | sr          | lr          | day.of.year |
| twd         | 0.07053484  | -0.23748086 | -0.1407255  |
| pr          | -0.26731509 | 0.38456779  | 0.0333563   |
| at          | 0.68725381  | 0.69517075  | 0.1750733   |
| ws          | -0.15772881 | -0.10127215 | -0.2133287  |
| dp          | 0.42904050  | 0.84979791  | 0.3075513   |
| sr          | 1.00000000  | 0.06950637  | -0.1729543  |
| lr          | 0.06950637  | 1.00000000  | 0.3402204   |
| day.of.year | -0.17295428 | 0.34022041  | 1.0000000   |

[1] "Picea abies"

|             |             |             |             |             |            |
|-------------|-------------|-------------|-------------|-------------|------------|
|             | twd         | pr          | at          | ws          | dp         |
| twd         | 1.00000000  | -0.11592144 | -0.25278882 | -0.02901937 | -0.3240365 |
| pr          | -0.11592144 | 1.00000000  | 0.01610968  | 0.09792613  | 0.1832309  |
| at          | -0.25278882 | 0.01610968  | 1.00000000  | 0.18377905  | 0.9194638  |
| ws          | -0.02901937 | 0.09792613  | 0.18377905  | 1.00000000  | 0.1295812  |
| dp          | -0.32403648 | 0.18323095  | 0.91946383  | 0.12958120  | 1.0000000  |
| sr          | -0.08383383 | -0.26611184 | 0.55808405  | 0.06462908  | 0.3973794  |
| lr          | -0.29126816 | 0.36264903  | 0.76861616  | 0.19288153  | 0.8906218  |
| day.of.year | 0.09163775  | 0.02764386  | 0.22703349  | -0.08269904 | 0.2947178  |

|     |             |            |             |
|-----|-------------|------------|-------------|
|     | sr          | lr         | day.of.year |
| twd | -0.08383383 | -0.2912682 | 0.09163775  |
| pr  | -0.26611184 | 0.3626490  | 0.02764386  |

|             |             |            |             |  |  |
|-------------|-------------|------------|-------------|--|--|
| at          | 0.55808405  | 0.7686162  | 0.22703349  |  |  |
| ws          | 0.06462908  | 0.1928815  | -0.08269904 |  |  |
| dp          | 0.39737939  | 0.8906218  | 0.29471780  |  |  |
| sr          | 1.00000000  | 0.1573929  | -0.19097898 |  |  |
| lr          | 0.15739289  | 1.00000000 | 0.25129408  |  |  |
| day.of.year | -0.19097898 | 0.2512941  | 1.00000000  |  |  |

[1] "Pinus sylvestris"

|             |             |             |             |              |              |
|-------------|-------------|-------------|-------------|--------------|--------------|
|             | twd         | pr          | at          | ws           | dp           |
| twd         | 1.00000000  | -0.17198315 | -0.01632060 | 0.030783564  | -0.152361842 |
| pr          | -0.17198315 | 1.00000000  | -0.00797577 | -0.065317673 | 0.202258877  |
| at          | -0.01632060 | -0.00797577 | 1.00000000  | 0.200675846  | 0.886176720  |
| ws          | 0.03078356  | -0.06531767 | 0.20067585  | 1.000000000  | 0.000612399  |
| dp          | -0.15236184 | 0.20225888  | 0.88617672  | 0.000612399  | 1.000000000  |
| sr          | 0.10575641  | -0.24077228 | 0.74514516  | 0.284416006  | 0.492799530  |
| lr          | -0.16483295 | 0.32570466  | 0.81373988  | 0.108458914  | 0.910680491  |
| day.of.year | -0.09338251 | 0.02089943  | 0.16894938  | -0.243834125 | 0.291081706  |

|             |            |            |             |
|-------------|------------|------------|-------------|
|             | sr         | lr         | day.of.year |
| twd         | 0.1057564  | -0.1648329 | -0.09338251 |
| pr          | -0.2407723 | 0.3257047  | 0.02089943  |
| at          | 0.7451452  | 0.8137399  | 0.16894938  |
| ws          | 0.2844160  | 0.1084589  | -0.24383412 |
| dp          | 0.4927995  | 0.9106805  | 0.29108171  |
| sr          | 1.0000000  | 0.3899210  | -0.15308436 |
| lr          | 0.3899210  | 1.0000000  | 0.23804080  |
| day.of.year | -0.1530844 | 0.2380408  | 1.00000000  |

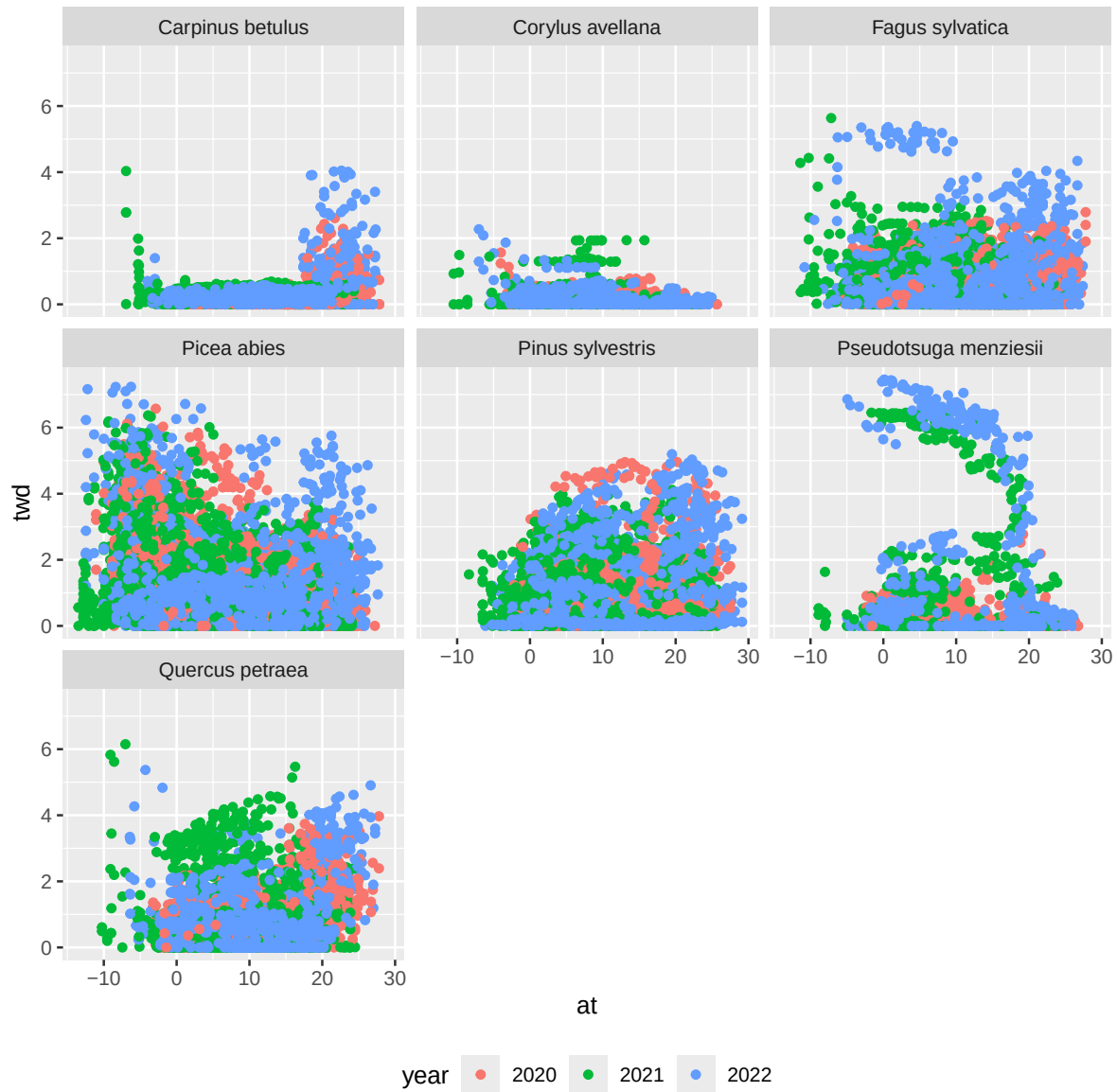
[1] "Pseudotsuga menziesii"

|             |             |              |             |             |             |
|-------------|-------------|--------------|-------------|-------------|-------------|
|             | twd         | pr           | at          | ws          | dp          |
| twd         | 1.00000000  | -0.028627835 | -0.16211835 | -0.07723890 | -0.08096724 |
| pr          | -0.02862784 | 1.000000000  | 0.05674352  | 0.28023423  | 0.14857411  |
| at          | -0.16211835 | 0.056743523  | 1.00000000  | -0.02143595 | 0.94181408  |
| ws          | -0.07723890 | 0.280234234  | -0.02143595 | 1.00000000  | -0.13053257 |
| dp          | -0.08096724 | 0.148574107  | 0.94181408  | -0.13053257 | 1.00000000  |
| sr          | -0.28368267 | -0.245028300 | 0.67765399  | 0.02164081  | 0.46844137  |
| lr          | -0.02895595 | 0.348268450  | 0.75399104  | 0.01009687  | 0.86082813  |
| day.of.year | 0.35193517  | 0.007705166  | 0.17007220  | -0.23853848 | 0.30313531  |

|     |             |             |              |
|-----|-------------|-------------|--------------|
|     | sr          | lr          | day.of.year  |
| twd | -0.28368267 | -0.02895595 | 0.351935169  |
| pr  | -0.24502830 | 0.34826845  | 0.007705166  |
| at  | 0.67765399  | 0.75399104  | 0.170072196  |
| ws  | 0.02164081  | 0.01009687  | -0.238538479 |
| dp  | 0.46844137  | 0.86082813  | 0.303135309  |
| sr  | 1.00000000  | 0.15082065  | -0.177108728 |
| lr  | 0.15082065  | 1.00000000  | 0.326631339  |

day.of.year -0.17710873 0.32663134 1.000000000

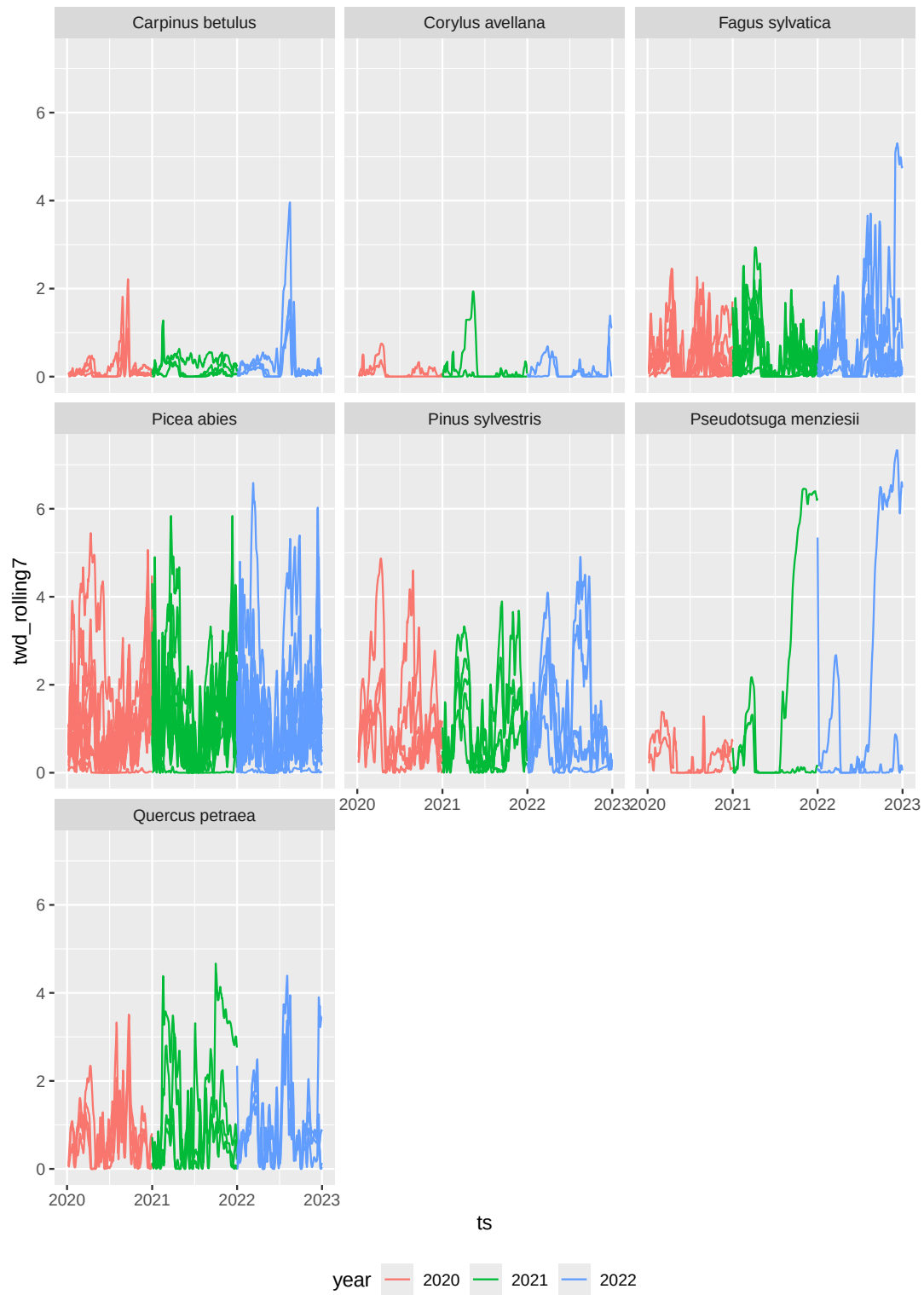
```
ggplot(  
  df.ts.series2,  
  aes(x = at, y = twd,  
      group = interaction(year, tree.id))) +  
  geom_point(aes(color = year)) +  
  facet_wrap(~species) +  
  theme(legend.position='bottom')
```



```
ggplot(  
  df.ts.series2,  
  aes(x = ts, y = twd_rolling7,  
    group = interaction(year, tree.id))) +  
  geom_line(aes(color = year)) +  
  facet_wrap(~species) +  
  theme(legend.position='bottom')
```

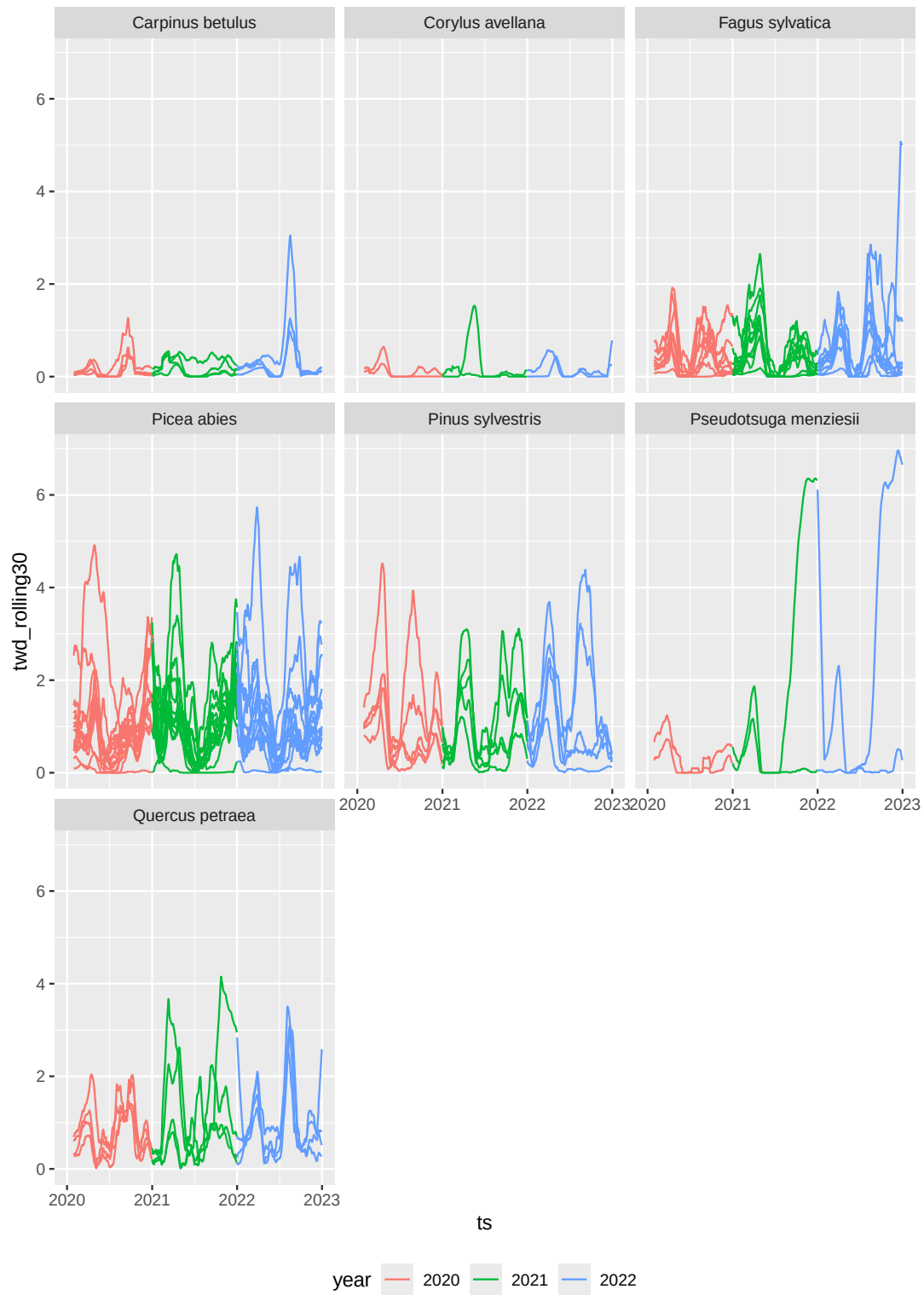
Warning: Removed 252 rows containing missing values or values outside the scale range (`geom\_line()`).





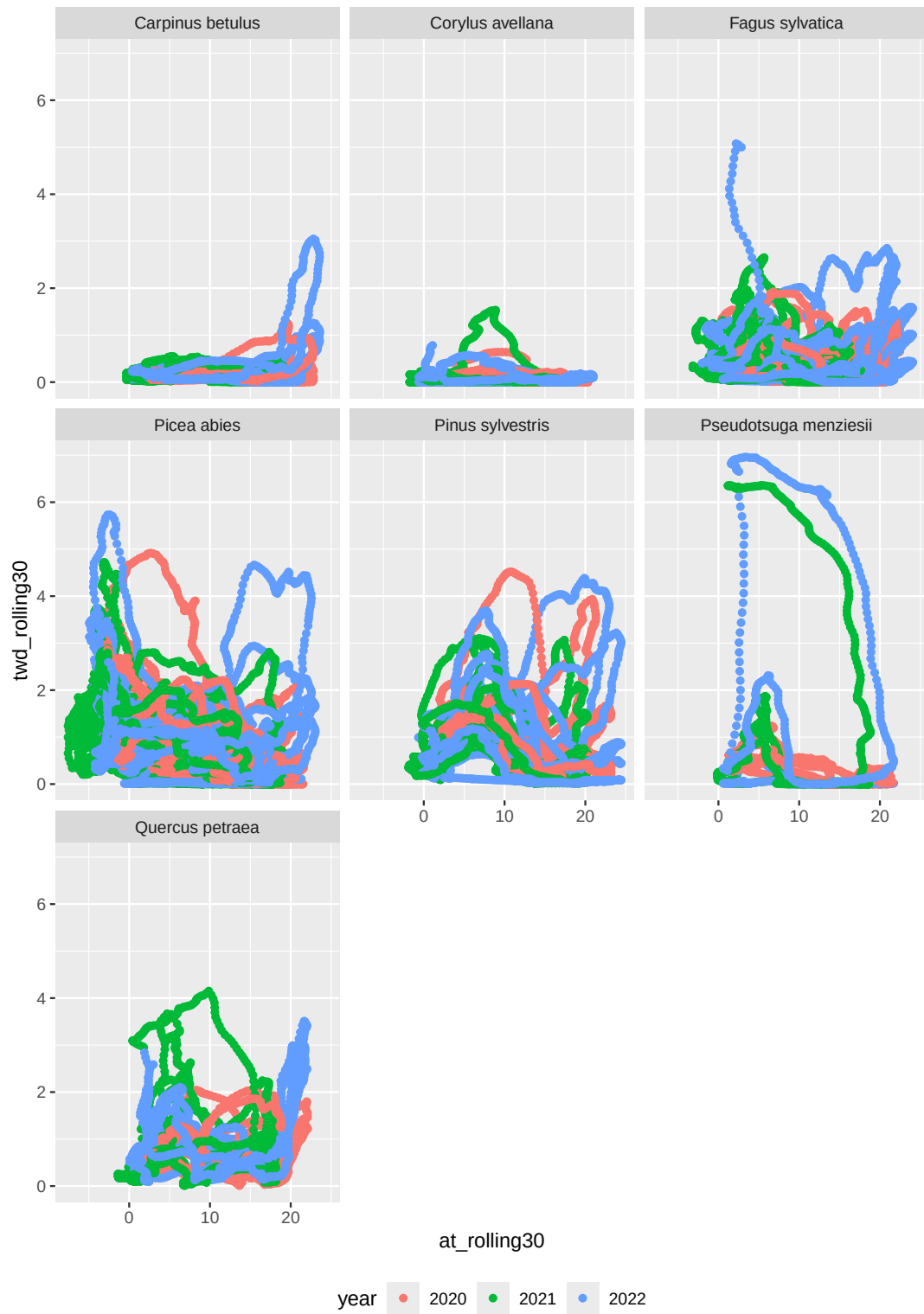
```
ggplot(  
  df.ts.series2,  
  aes(x = ts, y = twd_rolling30,  
    group = interaction(year, tree.id))) +  
  geom_line(aes(color = year)) +  
  #geom_point(aes(color = year), size = 1)+  
  facet_wrap(~species) +  
  theme(legend.position='bottom')
```

Warning: Removed 1218 rows containing missing values or values outside the scale range (`geom\_line()`).



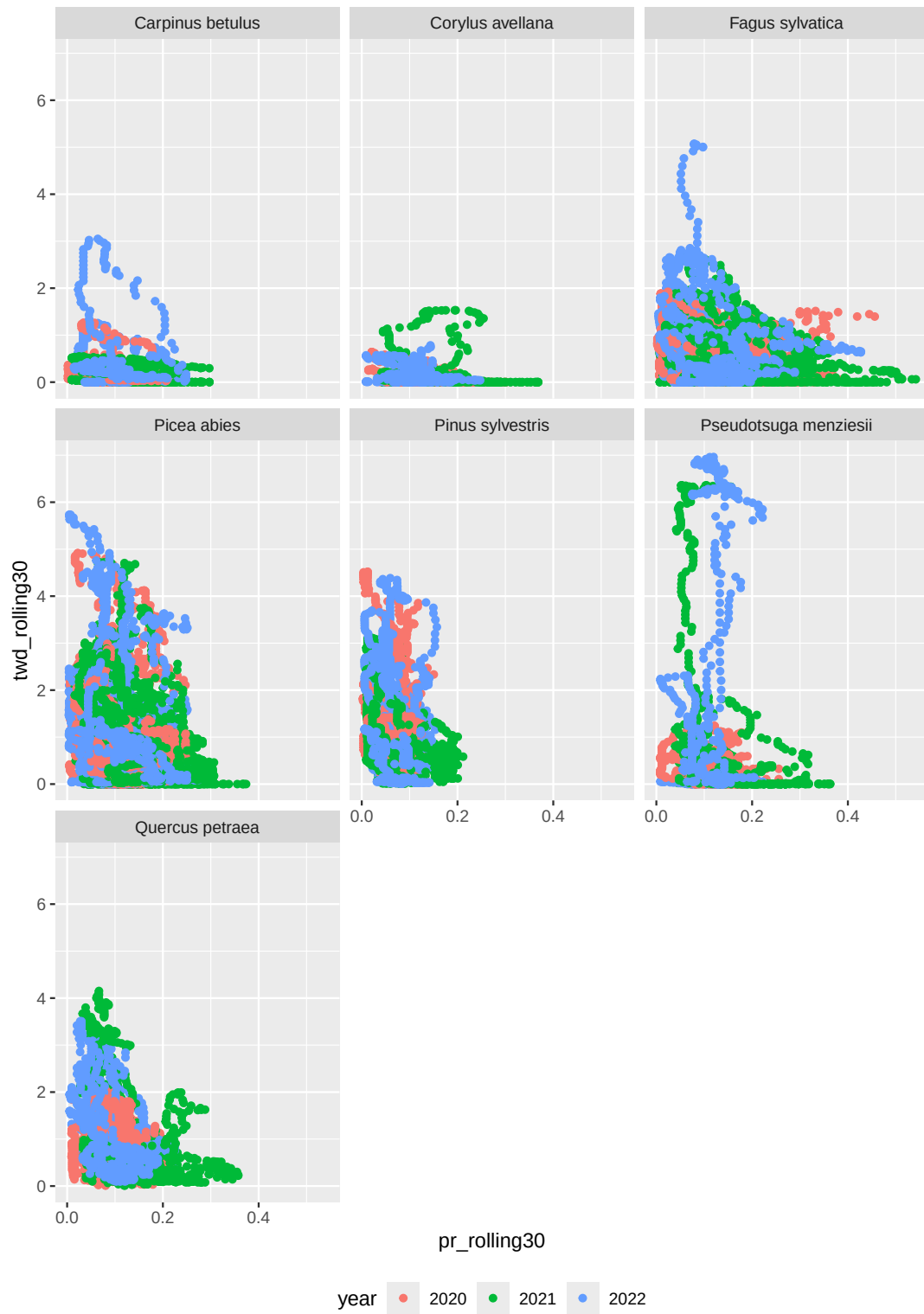
```
ggplot(  
  df.ts.series2,  
  aes(x = at_rolling30, y = twd_rolling30,  
    group = interaction(year, tree.id))) +  
  geom_point(aes(color = year)) +  
  facet_wrap(~species) +  
  theme(legend.position='bottom')
```

Warning: Removed 1218 rows containing missing values or values outside the scale range (`geom_point()`).



```
ggplot(  
  df.ts.series2,  
  aes(x = pr_rolling30, y = twd_rolling30,  
    group = interaction(year, tree.id))) +  
  geom_point(aes(color = year)) +  
  facet_wrap(~species) +  
  theme(legend.position='bottom')
```

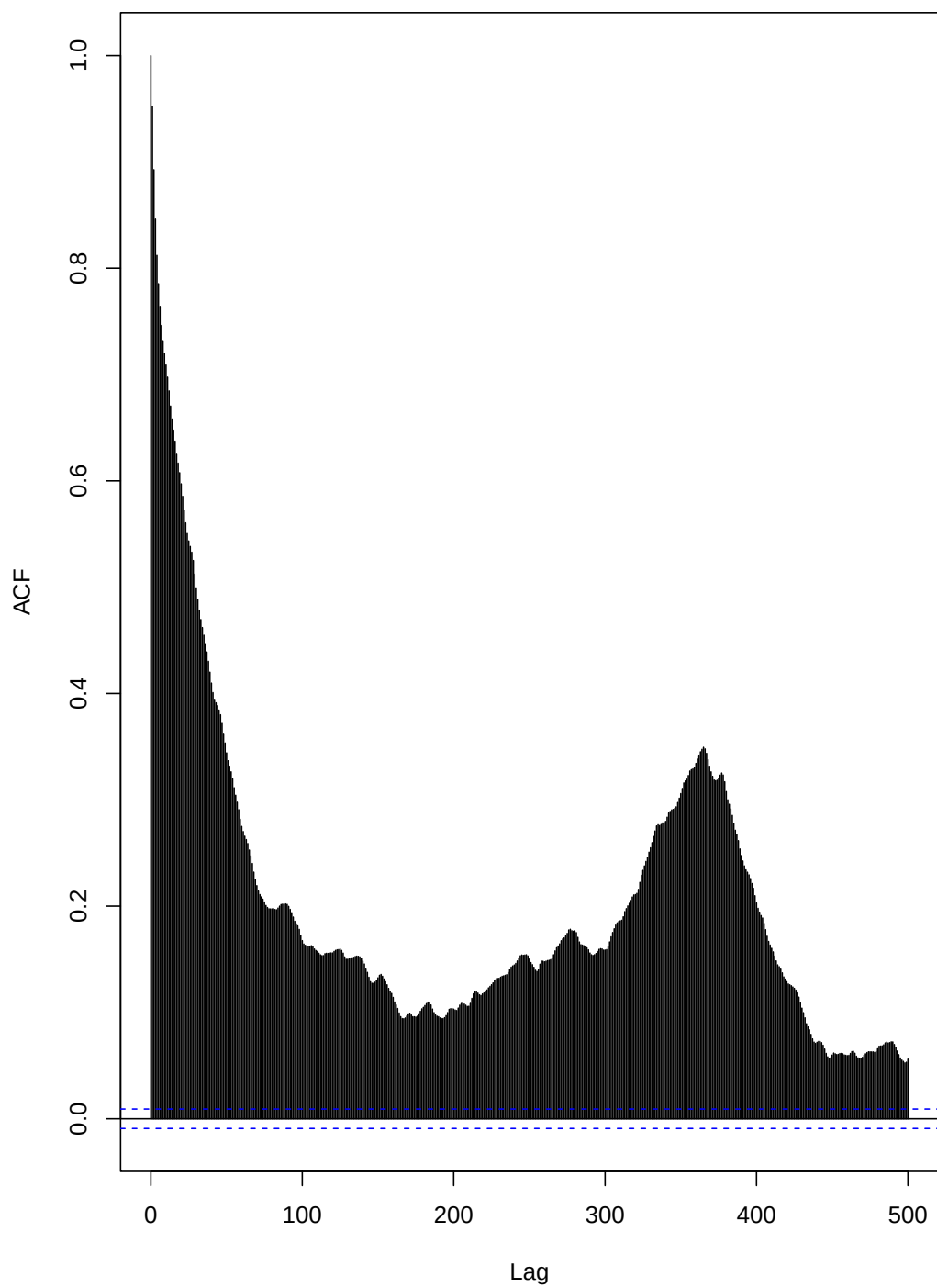
Warning: Removed 1218 rows containing missing values or values outside the scale range (``geom_point()``).



```
acf(df.ts.series2$twd, lag.max = 500)
```

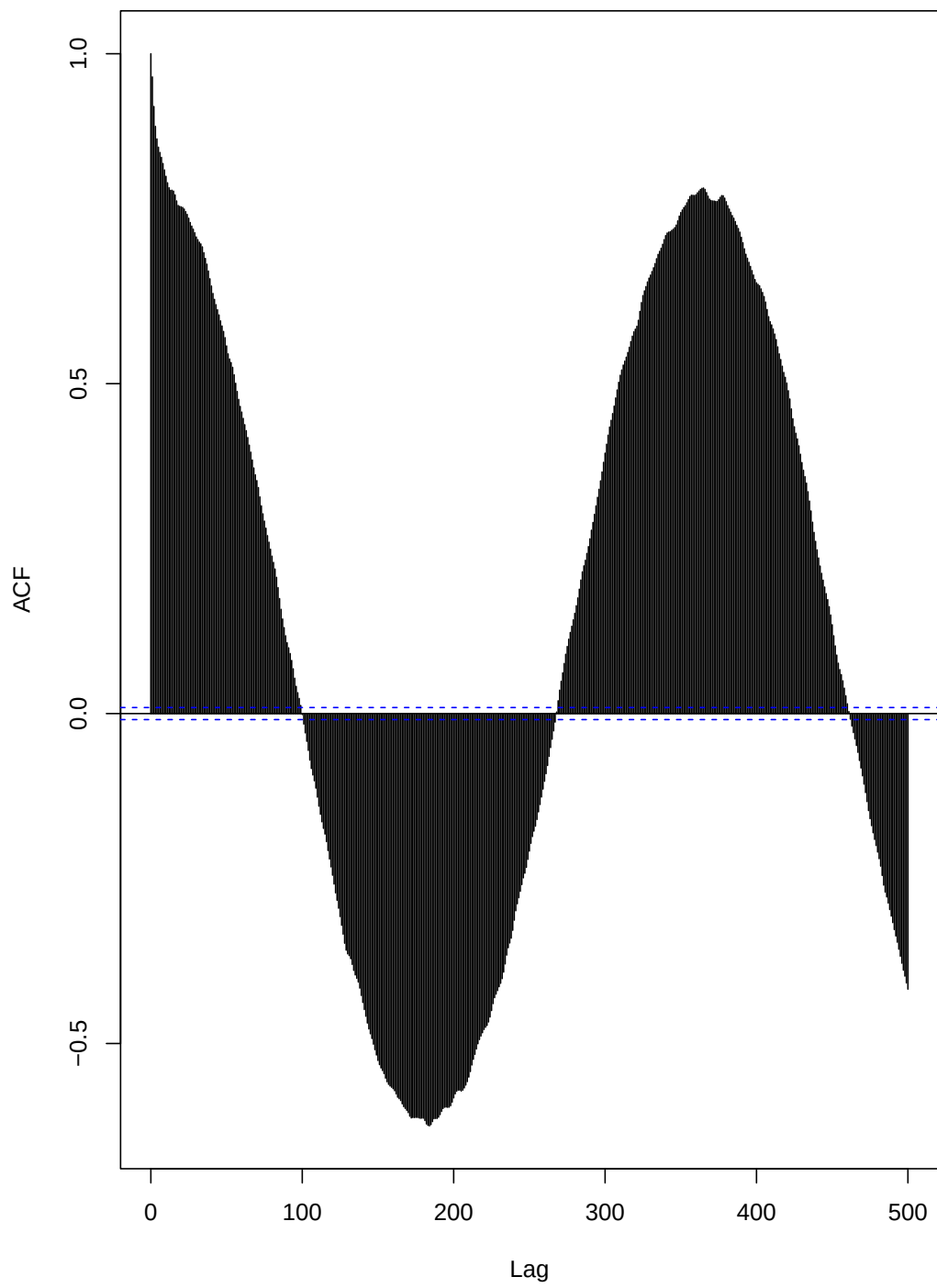


Series df.ts.series2\$twd



```
# clear cycl because high correlation again at 365 days  
acf(df.ts.series2$at, lag.max = 500)
```

Series df.ts.series2\$at



```

rolling.cols <- paste0(num.cols, "_rolling7")

for(iter_species in list_species){
  iter_df <- df.ts.series2 %>%
    filter(species == iter_species)

  print(iter_species)
  print(cor(iter_df[,rolling.cols], use = "complete.obs"))
}

```

[1] "Fagus sylvatica"

|                      | twd_rolling7 | pr_rolling7  | at_rolling7  | ws_rolling7  |
|----------------------|--------------|--------------|--------------|--------------|
| twd_rolling7         | 1.000000000  | -0.229326842 | -0.001806405 | 0.005359136  |
| pr_rolling7          | -0.229326842 | 1.000000000  | -0.002580239 | 0.056071004  |
| at_rolling7          | -0.001806405 | -0.002580239 | 1.000000000  | -0.054980460 |
| ws_rolling7          | 0.005359136  | 0.056071004  | -0.054980460 | 1.000000000  |
| dp_rolling7          | -0.155296191 | 0.152297003  | 0.927948643  | -0.169831250 |
| sr_rolling7          | 0.114709687  | -0.175259465 | 0.771449126  | 0.145150365  |
| lr_rolling7          | -0.190929474 | 0.316380407  | 0.834238844  | -0.108341723 |
| day.of.year_rolling7 | -0.057689827 | 0.049836083  | 0.188586244  | -0.343576437 |

|                      | dp_rolling7 | sr_rolling7 | lr_rolling7 | day.of.year_rolling7 |
|----------------------|-------------|-------------|-------------|----------------------|
| twd_rolling7         | -0.1552962  | 0.1147097   | -0.1909295  | -0.05768983          |
| pr_rolling7          | 0.1522970   | -0.1752595  | 0.3163804   | 0.04983608           |
| at_rolling7          | 0.9279486   | 0.7714491   | 0.8342388   | 0.18858624           |
| ws_rolling7          | -0.1698312  | 0.1451504   | -0.1083417  | -0.34357644          |
| dp_rolling7          | 1.0000000   | 0.5658505   | 0.9292750   | 0.32569578           |
| sr_rolling7          | 0.5658505   | 1.0000000   | 0.4319584   | -0.18819698          |
| lr_rolling7          | 0.9292750   | 0.4319584   | 1.0000000   | 0.34110600           |
| day.of.year_rolling7 | 0.3256958   | -0.1881970  | 0.3411060   | 1.00000000           |

[1] "Quercus petraea"

|                      | twd_rolling7 | pr_rolling7 | at_rolling7 | ws_rolling7 |
|----------------------|--------------|-------------|-------------|-------------|
| twd_rolling7         | 1.000000000  | -0.30638524 | 0.1291794   | -0.06552551 |
| pr_rolling7          | -0.30638524  | 1.000000000 | 0.1019861   | 0.17175190  |
| at_rolling7          | 0.12917941   | 0.10198611  | 1.00000000  | -0.19133876 |
| ws_rolling7          | -0.06552551  | 0.17175190  | -0.1913388  | 1.000000000 |
| dp_rolling7          | 0.03770180   | 0.23915998  | 0.9337133   | -0.27314040 |
| sr_rolling7          | 0.13450279   | -0.08281823 | 0.7746494   | -0.04617898 |
| lr_rolling7          | -0.03377644  | 0.39795395  | 0.8283397   | -0.14520137 |
| day.of.year_rolling7 | 0.08436441   | 0.07337225  | 0.1802469   | -0.40016896 |

|              | dp_rolling7 | sr_rolling7 | lr_rolling7 | day.of.year_rolling7 |
|--------------|-------------|-------------|-------------|----------------------|
| twd_rolling7 | 0.0377018   | 0.13450279  | -0.03377644 | 0.08436441           |
| pr_rolling7  | 0.2391600   | -0.08281823 | 0.39795395  | 0.07337225           |

|                      |            |             |             |             |
|----------------------|------------|-------------|-------------|-------------|
| at_rolling7          | 0.9337133  | 0.77464945  | 0.82833967  | 0.18024690  |
| ws_rolling7          | -0.2731404 | -0.04617898 | -0.14520137 | -0.40016896 |
| dp_rolling7          | 1.0000000  | 0.57116163  | 0.92560242  | 0.33106272  |
| sr_rolling7          | 0.5711616  | 1.00000000  | 0.40917217  | -0.18825438 |
| lr_rolling7          | 0.9256024  | 0.40917217  | 1.00000000  | 0.36296281  |
| day.of.year_rolling7 | 0.3310627  | -0.18825438 | 0.36296281  | 1.00000000  |

[1] "Carpinus betulus"

|                      |              |             |             |             |
|----------------------|--------------|-------------|-------------|-------------|
|                      | twd_rolling7 | pr_rolling7 | at_rolling7 | ws_rolling7 |
| twd_rolling7         | 1.00000000   | -0.18378950 | 0.23034814  | 0.1080840   |
| pr_rolling7          | -0.18378950  | 1.00000000  | -0.03441214 | -0.1183427  |
| at_rolling7          | 0.23034814   | -0.03441214 | 1.00000000  | -0.1927280  |
| ws_rolling7          | 0.10808400   | -0.11834274 | -0.19272802 | 1.0000000   |
| dp_rolling7          | 0.11815185   | 0.16405094  | 0.92631633  | -0.3276740  |
| sr_rolling7          | 0.21861630   | -0.30876256 | 0.78498848  | 0.1226974   |
| lr_rolling7          | 0.10235193   | 0.34682046  | 0.84670696  | -0.2890144  |
| day.of.year_rolling7 | 0.01473794   | 0.11614731  | 0.18483807  | -0.2857379  |

|                      |             |             |             |                      |
|----------------------|-------------|-------------|-------------|----------------------|
|                      | dp_rolling7 | sr_rolling7 | lr_rolling7 | day.of.year_rolling7 |
| twd_rolling7         | 0.1181519   | 0.2186163   | 0.1023519   | 0.01473794           |
| pr_rolling7          | 0.1640509   | -0.3087626  | 0.3468205   | 0.11614731           |
| at_rolling7          | 0.9263163   | 0.7849885   | 0.8467070   | 0.18483807           |
| ws_rolling7          | -0.3276740  | 0.1226974   | -0.2890144  | -0.28573789          |
| dp_rolling7          | 1.0000000   | 0.5617652   | 0.9263391   | 0.34582198           |
| sr_rolling7          | 0.5617652   | 1.0000000   | 0.4493856   | -0.18602970          |
| lr_rolling7          | 0.9263391   | 0.4493856   | 1.0000000   | 0.33152721           |
| day.of.year_rolling7 | 0.3458220   | -0.1860297  | 0.3315272   | 1.00000000           |

[1] "Corylus avellana"

|                      |              |             |             |             |
|----------------------|--------------|-------------|-------------|-------------|
|                      | twd_rolling7 | pr_rolling7 | at_rolling7 | ws_rolling7 |
| twd_rolling7         | 1.00000000   | -0.07885975 | -0.1682110  | 0.1546851   |
| pr_rolling7          | -0.07885975  | 1.00000000  | 0.1019305   | 0.1105735   |
| at_rolling7          | -0.16821102  | 0.10193045  | 1.0000000   | -0.3865053  |
| ws_rolling7          | 0.15468509   | 0.11057350  | -0.3865053  | 1.0000000   |
| dp_rolling7          | -0.27561466  | 0.23804869  | 0.9410117   | -0.4252089  |
| sr_rolling7          | 0.07162170   | -0.11374353 | 0.7803831   | -0.2179102  |
| lr_rolling7          | -0.28348277  | 0.43425538  | 0.7940991   | -0.3274269  |
| day.of.year_rolling7 | -0.16630779  | 0.05128859  | 0.1808164   | -0.3781703  |

|              |             |             |             |                      |
|--------------|-------------|-------------|-------------|----------------------|
|              | dp_rolling7 | sr_rolling7 | lr_rolling7 | day.of.year_rolling7 |
| twd_rolling7 | -0.2756147  | 0.0716217   | -0.2834828  | -0.16630779          |
| pr_rolling7  | 0.2380487   | -0.1137435  | 0.4342554   | 0.05128859           |
| at_rolling7  | 0.9410117   | 0.7803831   | 0.7940991   | 0.18081642           |
| ws_rolling7  | -0.4252089  | -0.2179102  | -0.3274269  | -0.37817030          |
| dp_rolling7  | 1.0000000   | 0.5654600   | 0.9164353   | 0.32683459           |
| sr_rolling7  | 0.5654600   | 1.0000000   | 0.3352121   | -0.20001558          |
| lr_rolling7  | 0.9164353   | 0.3352121   | 1.0000000   | 0.41194148           |

```

day.of.year_rolling7  0.3268346 -0.2000156  0.4119415          1.00000000
[1] "Picea abies"
      twd_rolling7 pr_rolling7 at_rolling7 ws_rolling7
twd_rolling7      1.00000000 -0.20874054 -0.2996081 -0.04517938
pr_rolling7      -0.20874054  1.00000000  0.1110258  0.11953581
at_rolling7      -0.29960813  0.11102580  1.0000000  0.22993358
ws_rolling7      -0.04517938  0.11953581  0.2299336  1.00000000
dp_rolling7      -0.35866377  0.26686138  0.9510824  0.17403626
sr_rolling7      -0.13222135 -0.07206777  0.6398348  0.11645644
lr_rolling7      -0.35930200  0.40184668  0.8844347  0.22414563
day.of.year_rolling7  0.08499879  0.04796793  0.2404778 -0.10711066
      dp_rolling7 sr_rolling7 lr_rolling7 day.of.year_rolling7
twd_rolling7      -0.3586638 -0.13222135 -0.3593020  0.08499879
pr_rolling7      0.2668614 -0.07206777  0.4018467  0.04796793
at_rolling7      0.9510824  0.63983482  0.8844347  0.24047781
ws_rolling7      0.1740363  0.11645644  0.2241456 -0.10711066
dp_rolling7      1.0000000  0.54566641  0.9468367  0.31473699
sr_rolling7      0.5456664  1.00000000  0.4483337 -0.22695706
lr_rolling7      0.9468367  0.44833369  1.0000000  0.29285342
day.of.year_rolling7  0.3147370 -0.22695706  0.2928534  1.00000000
[1] "Pinus sylvestris"
      twd_rolling7 pr_rolling7 at_rolling7 ws_rolling7
twd_rolling7      1.00000000 -0.27333886 -0.04136144  0.02710506
pr_rolling7      -0.27333886  1.00000000  0.04456262 -0.02295181
at_rolling7      -0.04136144  0.04456262  1.00000000  0.30524248
ws_rolling7      0.02710506 -0.02295181  0.30524248  1.00000000
dp_rolling7      -0.15757589  0.22765296  0.92877872  0.10103160
sr_rolling7      0.09045519 -0.09084960  0.81564605  0.45189083
lr_rolling7      -0.18601458  0.32015533  0.90516003  0.21152998
day.of.year_rolling7 -0.10346325  0.02338364  0.17750497 -0.40201000
      dp_rolling7 sr_rolling7 lr_rolling7 day.of.year_rolling7
twd_rolling7      -0.1575759  0.09045519 -0.1860146 -0.10346325
pr_rolling7      0.2276530 -0.09084960  0.3201553  0.02338364
at_rolling7      0.9287787  0.81564605  0.9051600  0.17750497
ws_rolling7      0.1010316  0.45189083  0.2115300 -0.40201000
dp_rolling7      1.0000000  0.63756182  0.9520371  0.31554406
sr_rolling7      0.6375618  1.00000000  0.6294887 -0.17071344
lr_rolling7      0.9520371  0.62948873  1.0000000  0.26915669
day.of.year_rolling7  0.3155441 -0.17071344  0.2691567  1.00000000
[1] "Pseudotsuga menziesii"
      twd_rolling7 pr_rolling7 at_rolling7 ws_rolling7
twd_rolling7      1.00000000 -0.069480132 -0.17336808 -0.13545554
pr_rolling7      -0.06948013  1.000000000  0.15756877  0.30883038

```

|                      |             |              |             |                      |
|----------------------|-------------|--------------|-------------|----------------------|
| at_rolling7          | -0.17336808 | 0.157568772  | 1.00000000  | -0.07111409          |
| ws_rolling7          | -0.13545554 | 0.308830377  | -0.07111409 | 1.00000000           |
| dp_rolling7          | -0.08813263 | 0.229368713  | 0.95678648  | -0.18813102          |
| sr_rolling7          | -0.32407623 | -0.039964409 | 0.77307368  | 0.08901946           |
| lr_rolling7          | -0.03583490 | 0.396940574  | 0.84393488  | -0.10810573          |
| day.of.year_rolling7 | 0.34901114  | 0.003027917  | 0.17559813  | -0.40367532          |
|                      | dp_rolling7 | sr_rolling7  | lr_rolling7 | day.of.year_rolling7 |
| twd_rolling7         | -0.08813263 | -0.32407623  | -0.0358349  | 0.349011137          |
| pr_rolling7          | 0.22936871  | -0.03996441  | 0.3969406   | 0.003027917          |
| at_rolling7          | 0.95678648  | 0.77307368   | 0.8439349   | 0.175598130          |
| ws_rolling7          | -0.18813102 | 0.08901946   | -0.1081057  | -0.403675317         |
| dp_rolling7          | 1.00000000  | 0.59765829   | 0.9268854   | 0.321053901          |
| sr_rolling7          | 0.59765829  | 1.00000000   | 0.4111937   | -0.206733747         |
| lr_rolling7          | 0.92688540  | 0.41119369   | 1.0000000   | 0.389889946          |
| day.of.year_rolling7 | 0.32105390  | -0.20673375  | 0.3898899   | 1.000000000          |

```
rolling.cols <- paste0(num.cols, "_rolling30")

for(iter_species in list_species){
  iter_df <- df.ts.series2 %>%
    filter(species == iter_species)

  print(iter_species)
  print(cor(iter_df[,rolling.cols], use = "complete.obs"))
}
```

[1] "Fagus sylvatica"

|                       | twd_rolling30 | pr_rolling30 | at_rolling30 | ws_rolling30 |
|-----------------------|---------------|--------------|--------------|--------------|
| twd_rolling30         | 1.00000000    | -0.212181423 | -0.03078646  | 0.018716165  |
| pr_rolling30          | -0.21218142   | 1.000000000  | 0.06315059   | 0.007965456  |
| at_rolling30          | -0.03078646   | 0.063150594  | 1.00000000   | -0.065010419 |
| ws_rolling30          | 0.01871617    | 0.007965456  | -0.06501042  | 1.000000000  |
| dp_rolling30          | -0.16949196   | 0.196012211  | 0.94343951   | -0.200585211 |
| sr_rolling30          | 0.08765254    | -0.038768951 | 0.81300739   | 0.195894994  |
| lr_rolling30          | -0.18927300   | 0.268092193  | 0.91127770   | -0.181026869 |
| day.of.year_rolling30 | -0.08776357   | 0.035581843  | 0.20287378   | -0.546695779 |
|                       | dp_rolling30  | sr_rolling30 | lr_rolling30 |              |
| twd_rolling30         | -0.1694920    | 0.08765254   | -0.1892730   |              |
| pr_rolling30          | 0.1960122     | -0.03876895  | 0.2680922    |              |
| at_rolling30          | 0.9434395     | 0.81300739   | 0.9112777    |              |
| ws_rolling30          | -0.2005852    | 0.19589499   | -0.1810269   |              |
| dp_rolling30          | 1.0000000     | 0.63619694   | 0.9689818    |              |

|                       |           |             |           |
|-----------------------|-----------|-------------|-----------|
| sr_rolling30          | 0.6361969 | 1.00000000  | 0.5902994 |
| lr_rolling30          | 0.9689818 | 0.59029936  | 1.0000000 |
| day.of.year_rolling30 | 0.3539579 | -0.20912885 | 0.3744793 |

day.of.year\_rolling30

|                       |             |
|-----------------------|-------------|
| twd_rolling30         | -0.08776357 |
| pr_rolling30          | 0.03558184  |
| at_rolling30          | 0.20287378  |
| ws_rolling30          | -0.54669578 |
| dp_rolling30          | 0.35395788  |
| sr_rolling30          | -0.20912885 |
| lr_rolling30          | 0.37447927  |
| day.of.year_rolling30 | 1.00000000  |

[1] "Quercus petraea"

|                       |               |              |              |              |
|-----------------------|---------------|--------------|--------------|--------------|
|                       | twd_rolling30 | pr_rolling30 | at_rolling30 | ws_rolling30 |
| twd_rolling30         | 1.000000000   | -0.36874741  | 0.1171246    | -0.04895769  |
| pr_rolling30          | -0.368747408  | 1.00000000   | 0.2419541    | 0.01102331   |
| at_rolling30          | 0.117124560   | 0.24195414   | 1.0000000    | -0.29375871  |
| ws_rolling30          | -0.048957693  | 0.01102331   | -0.2937587   | 1.00000000   |
| dp_rolling30          | 0.052358998   | 0.37015391   | 0.9450605    | -0.42147308  |
| sr_rolling30          | 0.093396270   | 0.13566319   | 0.8203822    | -0.01505648  |
| lr_rolling30          | 0.006242791   | 0.45023222   | 0.8987315    | -0.38354740  |
| day.of.year_rolling30 | 0.077171644   | 0.07828934   | 0.1944483    | -0.67546379  |

dp\_rolling30 sr\_rolling30 lr\_rolling30

|                       |            |             |              |
|-----------------------|------------|-------------|--------------|
| twd_rolling30         | 0.0523590  | 0.09339627  | 0.006242791  |
| pr_rolling30          | 0.3701539  | 0.13566319  | 0.450232220  |
| at_rolling30          | 0.9450605  | 0.82038215  | 0.898731476  |
| ws_rolling30          | -0.4214731 | -0.01505648 | -0.383547402 |
| dp_rolling30          | 1.0000000  | 0.64314346  | 0.967311019  |
| sr_rolling30          | 0.6431435  | 1.00000000  | 0.563999120  |
| lr_rolling30          | 0.9673110  | 0.56399912  | 1.000000000  |
| day.of.year_rolling30 | 0.3665343  | -0.21069044 | 0.409349735  |

day.of.year\_rolling30

|                       |             |
|-----------------------|-------------|
| twd_rolling30         | 0.07717164  |
| pr_rolling30          | 0.07828934  |
| at_rolling30          | 0.19444828  |
| ws_rolling30          | -0.67546379 |
| dp_rolling30          | 0.36653430  |
| sr_rolling30          | -0.21069044 |
| lr_rolling30          | 0.40934974  |
| day.of.year_rolling30 | 1.00000000  |

[1] "Carpinus betulus"

|               |               |              |              |              |
|---------------|---------------|--------------|--------------|--------------|
|               | twd_rolling30 | pr_rolling30 | at_rolling30 | ws_rolling30 |
| twd_rolling30 | 1.00000000    | -0.23788812  | 0.25038261   | 0.13044734   |



```

pr_rolling30          -0.23788812   1.00000000  -0.01315086  -0.08559449
at_rolling30          0.25038261  -0.01315086   1.00000000  -0.21996377
ws_rolling30          0.13044734  -0.08559449  -0.21996377   1.00000000
dp_rolling30          0.15077268   0.18238453   0.94397037  -0.37021966
sr_rolling30          0.21999877  -0.24996622   0.83255191   0.11796458
lr_rolling30          0.14878405   0.26446025   0.92790566  -0.31985430
day.of.year_rolling30 0.01098617   0.16153927   0.19685936  -0.52460399
      dp_rolling30 sr_rolling30 lr_rolling30
twd_rolling30         0.1507727    0.2199988    0.1487840
pr_rolling30          0.1823845   -0.2499662    0.2644602
at_rolling30          0.9439704    0.8325519    0.9279057
ws_rolling30         -0.3702197    0.1179646   -0.3198543
dp_rolling30          1.0000000    0.6565611    0.9643534
sr_rolling30          0.6565611    1.0000000    0.6376992
lr_rolling30          0.9643534    0.6376992    1.0000000
day.of.year_rolling30 0.3740399   -0.2038241    0.3547322
      day.of.year_rolling30
twd_rolling30         0.01098617
pr_rolling30          0.16153927
at_rolling30          0.19685936
ws_rolling30         -0.52460399
dp_rolling30          0.37403987
sr_rolling30         -0.20382410
lr_rolling30          0.35473221
day.of.year_rolling30 1.00000000
[1] "Corylus avellana"
      twd_rolling30 pr_rolling30 at_rolling30 ws_rolling30
twd_rolling30      1.0000000  -0.11878944  -0.1671747   0.2464720
pr_rolling30      -0.1187894   1.00000000   0.2465994  -0.0767915
at_rolling30      -0.1671747   0.24659940   1.0000000  -0.5759521
ws_rolling30       0.2464720  -0.07679150  -0.5759521   1.0000000
dp_rolling30      -0.2904390   0.35930843   0.9530411  -0.6438180
sr_rolling30       0.1081551   0.10711011   0.8222464  -0.2876951
lr_rolling30      -0.3301206   0.47807945   0.8707749  -0.6352241
day.of.year_rolling30 -0.2578483   0.02697776   0.1955792  -0.6281890
      dp_rolling30 sr_rolling30 lr_rolling30
twd_rolling30      -0.2904390   0.1081551  -0.3301206
pr_rolling30       0.3593084   0.1071101   0.4780794
at_rolling30       0.9530411   0.8222464   0.8707749
ws_rolling30      -0.6438180  -0.2876951  -0.6352241
dp_rolling30       1.0000000   0.6386810   0.9624457
sr_rolling30       0.6386810   1.0000000   0.4973503
lr_rolling30       0.9624457   0.4973503   1.0000000

```

```

day.of.year_rolling30    0.3603965   -0.2236869    0.4627945
                        day.of.year_rolling30
twd_rolling30           -0.25784831
pr_rolling30            0.02697776
at_rolling30            0.19557921
ws_rolling30           -0.62818899
dp_rolling30            0.36039651
sr_rolling30           -0.22368692
lr_rolling30            0.46279450
day.of.year_rolling30    1.00000000
[1] "Picea abies"

                        twd_rolling30 pr_rolling30 at_rolling30 ws_rolling30
twd_rolling30           1.00000000  -0.25361356  -0.3740038  -0.04551062
pr_rolling30           -0.25361356   1.00000000   0.2638959   0.08024032
at_rolling30           -0.37400381   0.26389586   1.0000000   0.26043912
ws_rolling30           -0.04551062   0.08024032   0.2604391   1.00000000
dp_rolling30           -0.41169556   0.39660243   0.9664684   0.19076659
sr_rolling30           -0.20270287   0.15082024   0.6686832   0.14211952
lr_rolling30           -0.41440303   0.43270205   0.9547638   0.22413467
day.of.year_rolling30    0.06133371   0.06301339   0.2580739  -0.12940025
                        dp_rolling30 sr_rolling30 lr_rolling30
twd_rolling30           -0.4116956  -0.2027029  -0.4144030
pr_rolling30            0.3966024   0.1508202   0.4327021
at_rolling30            0.9664684   0.6686832   0.9547638
ws_rolling30            0.1907666   0.1421195   0.2241347
dp_rolling30            1.0000000   0.6078743   0.9771511
sr_rolling30            0.6078743   1.0000000   0.5872998
lr_rolling30            0.9771511   0.5872998   1.0000000
day.of.year_rolling30    0.3339928  -0.2555050   0.3161751
                        day.of.year_rolling30
twd_rolling30           0.06133371
pr_rolling30           0.06301339
at_rolling30           0.25807386
ws_rolling30           -0.12940025
dp_rolling30           0.33399281
sr_rolling30           -0.25550496
lr_rolling30           0.31617506
day.of.year_rolling30    1.00000000
[1] "Pinus sylvestris"

                        twd_rolling30 pr_rolling30 at_rolling30 ws_rolling30
twd_rolling30           1.00000000  -0.33879900  -0.08690945  -0.008164029
pr_rolling30           -0.338799001  1.00000000   0.18429644   0.016437757
at_rolling30           -0.086909445   0.18429644   1.00000000   0.346168111

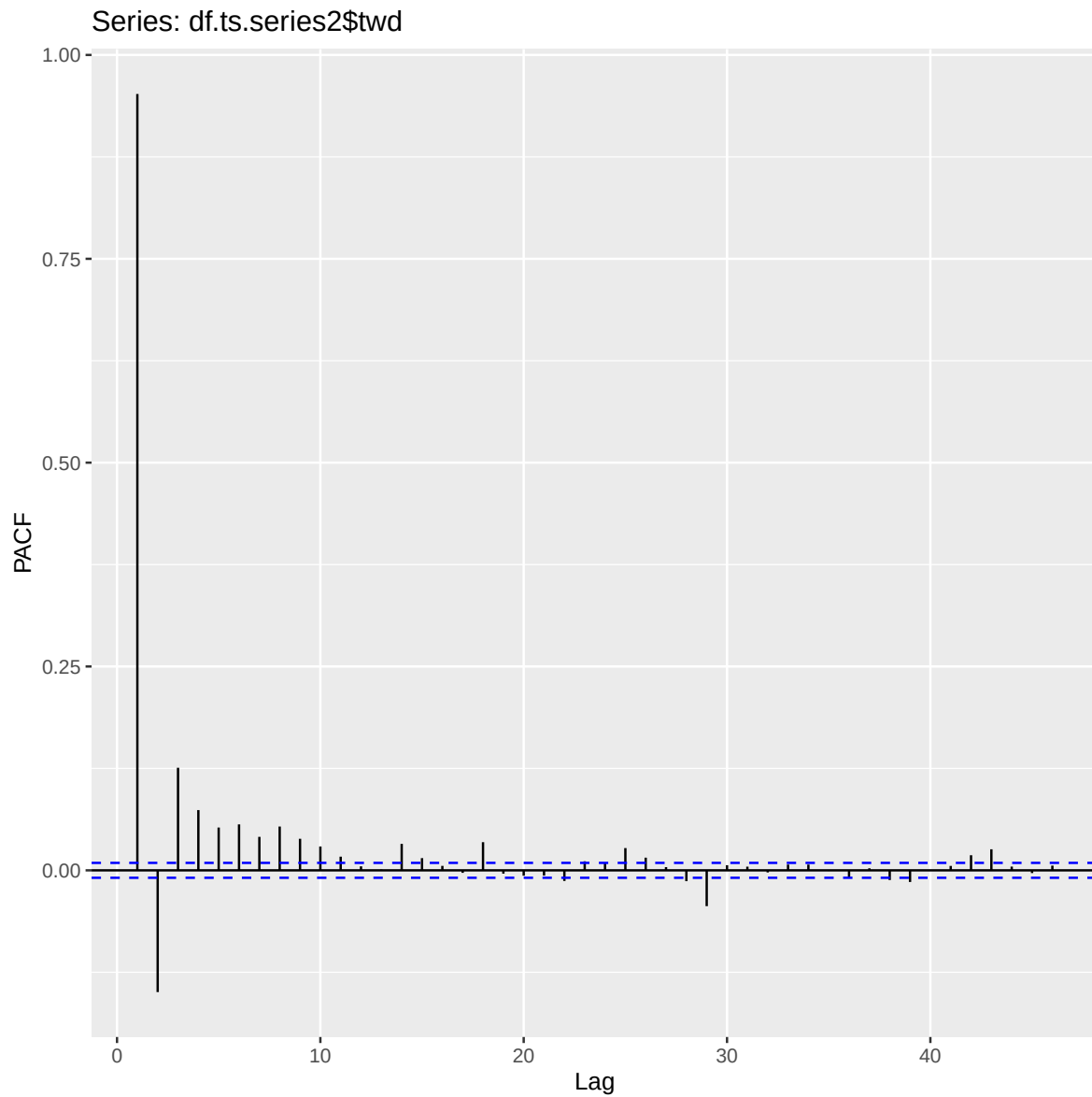
```

|                             |                       |              |              |              |
|-----------------------------|-----------------------|--------------|--------------|--------------|
| ws_rolling30                | -0.008164029          | 0.01643776   | 0.34616811   | 1.000000000  |
| dp_rolling30                | -0.167163341          | 0.31992692   | 0.95411170   | 0.146955607  |
| sr_rolling30                | 0.051004139           | 0.10062858   | 0.84566925   | 0.532112219  |
| lr_rolling30                | -0.205035562          | 0.35441665   | 0.96331873   | 0.258206033  |
| day.of.year_rolling30       | -0.116665051          | -0.03373590  | 0.18836405   | -0.552914814 |
|                             | dp_rolling30          | sr_rolling30 | lr_rolling30 |              |
| twd_rolling30               | -0.1671633            | 0.05100414   | -0.2050356   |              |
| pr_rolling30                | 0.3199269             | 0.10062858   | 0.3544166    |              |
| at_rolling30                | 0.9541117             | 0.84566925   | 0.9633187    |              |
| ws_rolling30                | 0.1469556             | 0.53211222   | 0.2582060    |              |
| dp_rolling30                | 1.0000000             | 0.70705369   | 0.9759720    |              |
| sr_rolling30                | 0.7070537             | 1.00000000   | 0.7461934    |              |
| lr_rolling30                | 0.9759720             | 0.74619340   | 1.0000000    |              |
| day.of.year_rolling30       | 0.3381932             | -0.18701580  | 0.2824684    |              |
|                             | day.of.year_rolling30 |              |              |              |
| twd_rolling30               | -0.1166651            |              |              |              |
| pr_rolling30                | -0.0337359            |              |              |              |
| at_rolling30                | 0.1883640             |              |              |              |
| ws_rolling30                | -0.5529148            |              |              |              |
| dp_rolling30                | 0.3381932             |              |              |              |
| sr_rolling30                | -0.1870158            |              |              |              |
| lr_rolling30                | 0.2824684             |              |              |              |
| day.of.year_rolling30       | 1.0000000             |              |              |              |
| [1] "Pseudotsuga menziesii" |                       |              |              |              |
|                             | twd_rolling30         | pr_rolling30 | at_rolling30 | ws_rolling30 |
| twd_rolling30               | 1.000000000           | -0.15828309  | -0.1792531   | -0.2135736   |
| pr_rolling30                | -0.15828309           | 1.000000000  | 0.3402710    | 0.2040199    |
| at_rolling30                | -0.17925312           | 0.34027104   | 1.0000000    | -0.1447002   |
| ws_rolling30                | -0.21357355           | 0.20401994   | -0.1447002   | 1.0000000    |
| dp_rolling30                | -0.08823155           | 0.38739268   | 0.9654277    | -0.2883100   |
| sr_rolling30                | -0.33875695           | 0.21909028   | 0.8164631    | 0.1727388    |
| lr_rolling30                | -0.04467526           | 0.48807141   | 0.9060974    | -0.3017098   |
| day.of.year_rolling30       | 0.33891713            | -0.04237303  | 0.1881446    | -0.6959699   |
|                             | dp_rolling30          | sr_rolling30 | lr_rolling30 |              |
| twd_rolling30               | -0.08823155           | -0.3387570   | -0.04467526  |              |
| pr_rolling30                | 0.38739268            | 0.2190903    | 0.48807141   |              |
| at_rolling30                | 0.96542774            | 0.8164631    | 0.90609741   |              |
| ws_rolling30                | -0.28830998           | 0.1727388    | -0.30170975  |              |
| dp_rolling30                | 1.00000000            | 0.6598778    | 0.97047002   |              |
| sr_rolling30                | 0.65987783            | 1.0000000    | 0.55425872   |              |
| lr_rolling30                | 0.97047002            | 0.5542587    | 1.00000000   |              |
| day.of.year_rolling30       | 0.35204798            | -0.2317087   | 0.43026229   |              |
|                             | day.of.year_rolling30 |              |              |              |

|                       |             |
|-----------------------|-------------|
| twd_rolling30         | 0.33891713  |
| pr_rolling30          | -0.04237303 |
| at_rolling30          | 0.18814460  |
| ws_rolling30          | -0.69596994 |
| dp_rolling30          | 0.35204798  |
| sr_rolling30          | -0.23170870 |
| lr_rolling30          | 0.43026229  |
| day.of.year_rolling30 | 1.00000000  |

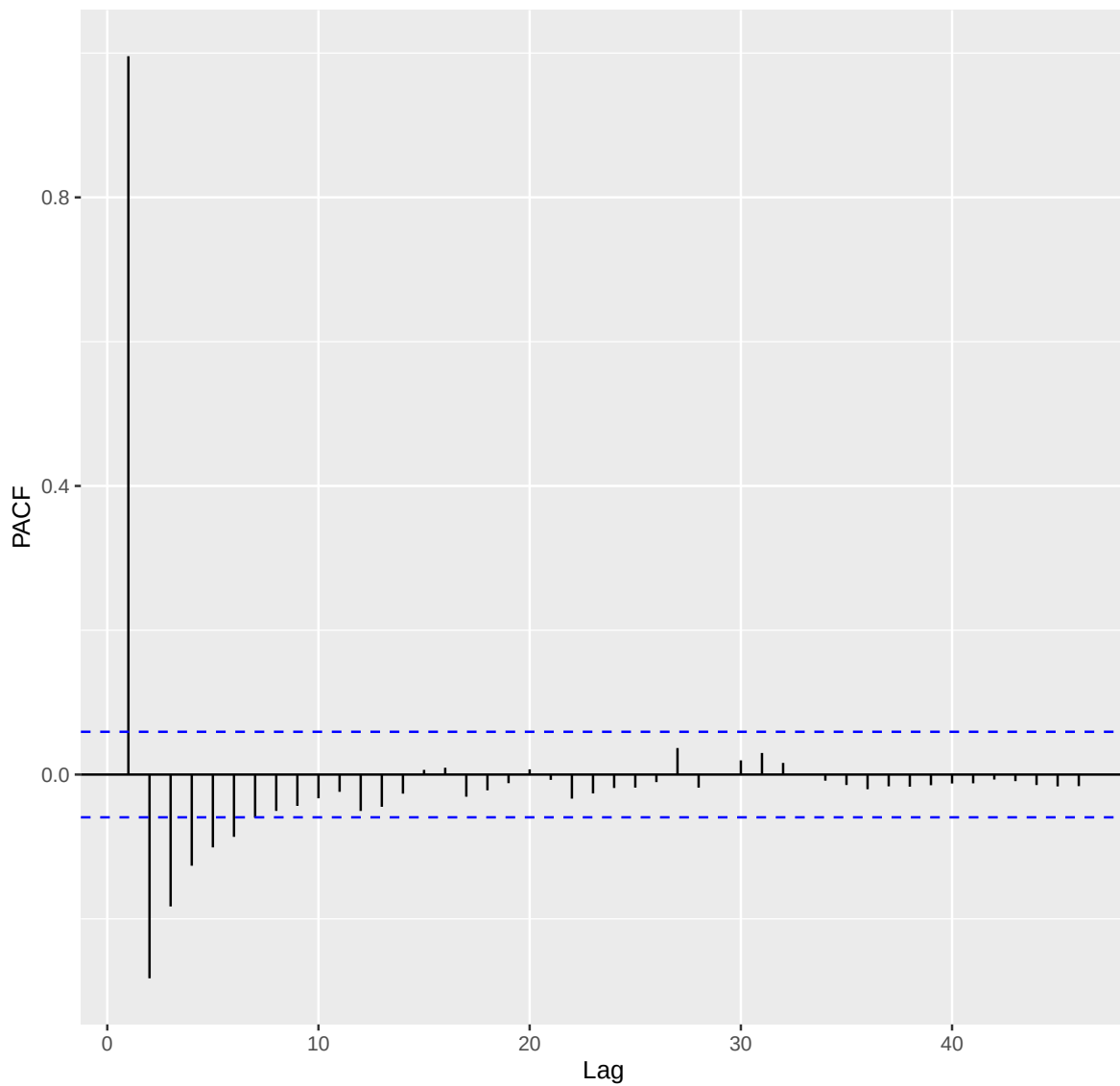
## Partial Autocorrelation

```
# Plot PACF  
ggPacf(df.ts.series2$twd)
```



```
ggPacf(df.ts.series2$twd_rolling30)
```

Series: df.ts.series2\$twd\_rolling30



```
plot_twd_lag <- function(df, x_col, twd_col, lag = 1) {  
  
  # Arrange data by time  
  df <- df %>% arrange(.data[[x_col]])  
  
  # Create lagged TWD values  
  df <- df %>%  
    mutate(Lagged_TWD = lag(.data[[twd_col]], lag))  
}
```

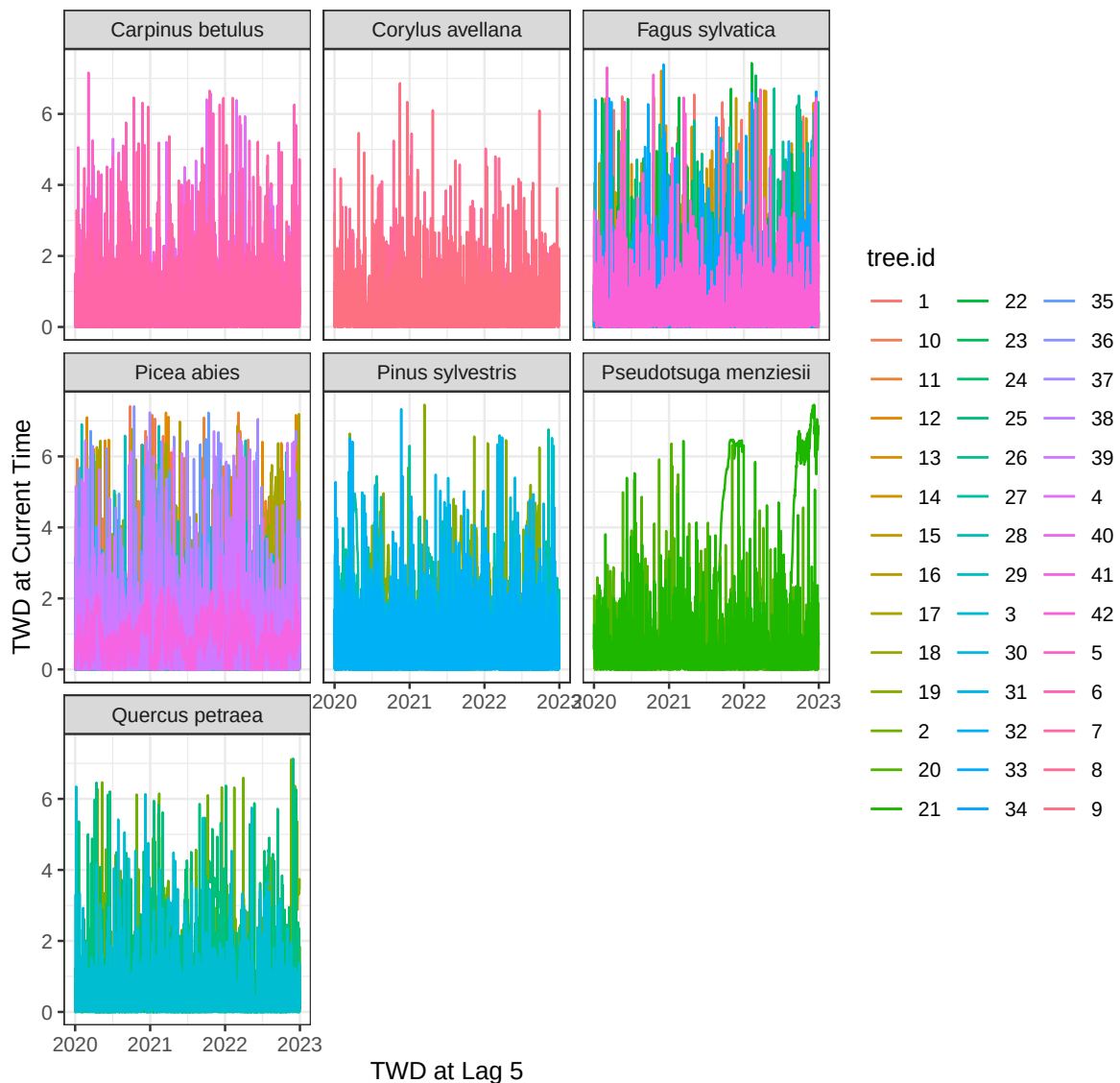
```

# Plot TWD vs. lagged TWD
ggplot(df, aes(x = ts, color = tree.id, y = .data[[twd_col]])) +
  geom_line() +
  geom_line(aes(y= Lagged_TWD))+
  labs(title = paste("Tree Water Deficit vs. Lagged TWD (Lag =", lag, ")"),
        x = paste("TWD at Lag", lag),
        y = "TWD at Current Time") +
  theme_bw() +
  facet_wrap(~species)
}

plot_twd_lag(df.ts.series2, "at", "twd", lag = 5)

```

## Tree Water Deficit vs. Lagged TWD (Lag = 5 )



## Granger Causality

Granger Causality: Rather than testing whether X causes Y, the Granger causality tests whether X forecasts Y.

A time series X is said to Granger-cause Y if it can be shown, usually through a series of t-tests and F-tests on lagged values of X (and with lagged values of Y also included), that those X values provide statistically significant information about future values of Y.



A variable X that evolves over time Granger-causes another evolving variable Y if predictions of the value of Y based on its own past values and on the past values of X are better than predictions of Y based only on Y's own past values.

If the variables are non-stationary, then the test is done using first (or higher) differences. The number of lags to be included is usually chosen using an information criterion

```
library(vars)
library(urca)
library(vars)

summary(ur.df(df.ts.series2$twd, type="drift", lags=2))
```

```
#####
# Augmented Dickey-Fuller Test Unit Root Test #
#####
```

Test regression drift

Call:

```
lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)
```

Residuals:

| Min     | 1Q      | Median  | 3Q     | Max    |
|---------|---------|---------|--------|--------|
| -6.3454 | -0.0568 | -0.0214 | 0.0910 | 5.0259 |

Coefficients:

|             | Estimate  | Std. Error | t value | Pr(> t )   |
|-------------|-----------|------------|---------|------------|
| (Intercept) | 0.035892  | 0.001730   | 20.74   | <2e-16 *** |
| z.lag.1     | -0.048081 | 0.001437   | -33.45  | <2e-16 *** |
| z.diff.lag1 | 0.161276  | 0.004593   | 35.11   | <2e-16 *** |
| z.diff.lag2 | -0.125885 | 0.004624   | -27.22  | <2e-16 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2913 on 46025 degrees of freedom

Multiple R-squared: 0.06084, Adjusted R-squared: 0.06077

F-statistic: 993.8 on 3 and 46025 DF, p-value: < 2.2e-16

Value of test-statistic is: -33.4549 559.6152

Critical values for test statistics:

|      | 1pct  | 5pct  | 10pct |
|------|-------|-------|-------|
| tau2 | -3.43 | -2.86 | -2.57 |
| phi1 | 6.43  | 4.59  | 3.78  |

```
summary(ur.df(df.ts.series2$at, type="drift", lags=2))
```

```
#####  
# Augmented Dickey-Fuller Test Unit Root Test #  
#####
```

Test regression drift

Call:

```
lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)
```

Residuals:

|  | Min      | 1Q      | Median | 3Q     | Max     |
|--|----------|---------|--------|--------|---------|
|  | -15.8620 | -1.1118 | 0.1609 | 1.2507 | 13.9444 |

Coefficients:

|             | Estimate  | Std. Error | t value | Pr(> t )   |
|-------------|-----------|------------|---------|------------|
| (Intercept) | 0.291496  | 0.014314   | 20.36   | <2e-16 *** |
| z.lag.1     | -0.032011 | 0.001206   | -26.54  | <2e-16 *** |
| z.diff.lag1 | 0.183176  | 0.004529   | 40.45   | <2e-16 *** |
| z.diff.lag2 | -0.212788 | 0.004554   | -46.72  | <2e-16 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.97 on 46025 degrees of freedom

Multiple R-squared: 0.08551, Adjusted R-squared: 0.08545

F-statistic: 1434 on 3 and 46025 DF, p-value: < 2.2e-16

Value of test-statistic is: -26.5441 352.2943

Critical values for test statistics:

|      | 1pct  | 5pct  | 10pct |
|------|-------|-------|-------|
| tau2 | -3.43 | -2.86 | -2.57 |

```
phi1 6.43 4.59 3.78
```

This tests the null hypothesis: The series has a unit root (i.e., non-stationary).

If the coefficient on `z.lag.1` is significantly different from zero (with a very small p-value), then you can reject the null hypothesis and conclude the series is stationary.

```
granger_test <- function(data, group = NA, x_col, y_col, lags = 10) {  
  run_test <- function(df_sub) {  
    df <- df_sub[, c(x_col, y_col)]  
    names(df) <- c("x", "y")  
    df <- df[complete.cases(df), ]  
  
    if (nrow(df) <= lags + 1) {  
      return("Not enough observations for VAR model.")  
    }  
  
    var_model <- VAR(df, p = lags, type = "const")  
    return(causality(var_model, cause = "x"))  
  }  
  
  if (is.na(group)) {  
    return(run_test(data))  
  } else {  
    group_list <- split(data, data[[group]])  
    results <- lapply(group_list, run_test)  
    return(results)  
  }  
}  
  
granger_test(df.ts.series2, y_col= "twd", x_col= "at", group = "tree.id")
```

```
$`1`  
$`1`$Granger
```

Granger causality H0: x do not Granger-cause y

```
data: VAR object var_model  
F-Test = 2.0838, df1 = 10, df2 = 2130, p-value = 0.02271
```

```
$`1`$Instant
```

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 15.657, df = 1, p-value = 7.592e-05

\$`10`

\$`10`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 0.80467, df1 = 10, df2 = 2130, p-value = 0.6243

\$`10`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 4.5935, df = 1, p-value = 0.03209

\$`11`

\$`11`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 4.2343, df1 = 10, df2 = 2130, p-value = 7.48e-06

\$`11`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 0.52412, df = 1, p-value = 0.4691

\$`12`

\$`12`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 7.6698, df1 = 10, df2 = 2130, p-value = 3.819e-12

\$`12`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 14.872, df = 1, p-value = 0.0001151

\$`13`

\$`13`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 1.8681, df1 = 10, df2 = 2130, p-value = 0.04516

\$`13`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 8.271, df = 1, p-value = 0.004028

\$`14`

\$`14`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 2.0877, df1 = 10, df2 = 2130, p-value = 0.02242

\$`14`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 1.6895, df = 1, p-value = 0.1937

\$`15`

\$`15`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 1.248, df1 = 10, df2 = 2130, p-value = 0.255

\$`15`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 1.7137, df = 1, p-value = 0.1905

\$`16`

\$`16`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 0.93515, df1 = 10, df2 = 2130, p-value = 0.4994

\$`16`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model  
Chi-squared = 6.3133, df = 1, p-value = 0.01198

\$`17`  
\$`17`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model  
F-Test = 1.1713, df1 = 10, df2 = 2130, p-value = 0.3055

\$`17`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model  
Chi-squared = 12.774, df = 1, p-value = 0.0003515

\$`18`  
\$`18`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model  
F-Test = 1.9908, df1 = 10, df2 = 2130, p-value = 0.03068

\$`18`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model  
Chi-squared = 5.2238, df = 1, p-value = 0.02228

\$`19`  
\$`19`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 2.0781, df1 = 10, df2 = 2130, p-value = 0.02314

\$`19`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 101.76, df = 1, p-value < 2.2e-16

\$`2`

\$`2`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 1.7357, df1 = 10, df2 = 2130, p-value = 0.0676

\$`2`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 20.325, df = 1, p-value = 6.533e-06

\$`20`

\$`20`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 1.9194, df1 = 10, df2 = 2130, p-value = 0.03848



\$`20`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 0.066187, df = 1, p-value = 0.797

\$`21`

\$`21`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 1.4587, df1 = 10, df2 = 2130, p-value = 0.1488

\$`21`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 16.61, df = 1, p-value = 4.591e-05

\$`22`

\$`22`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 2.9455, df1 = 10, df2 = 2130, p-value = 0.001111

\$`22`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 50.178, df = 1, p-value = 1.404e-12

\$`23`

\$`23`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 1.3749, df1 = 10, df2 = 2130, p-value = 0.1856

\$`23`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 10.015, df = 1, p-value = 0.001553

\$`24`

\$`24`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 2.3123, df1 = 10, df2 = 2130, p-value = 0.01059

\$`24`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 18.165, df = 1, p-value = 2.026e-05

\$`25`

\$`25`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model  
F-Test = 1.4335, df1 = 10, df2 = 2130, p-value = 0.1591

\$`25`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model  
Chi-squared = 44.199, df = 1, p-value = 2.966e-11

\$`26`

\$`26`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model  
F-Test = 2.924, df1 = 10, df2 = 2130, p-value = 0.001202

\$`26`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model  
Chi-squared = 40.808, df = 1, p-value = 1.679e-10

\$`27`

\$`27`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model  
F-Test = 2.2365, df1 = 10, df2 = 2130, p-value = 0.01369

\$`27`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model  
Chi-squared = 65.717, df = 1, p-value = 5.551e-16

\$`28`  
\$`28`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model  
F-Test = 1.0626, df1 = 10, df2 = 2130, p-value = 0.3879

\$`28`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model  
Chi-squared = 49.767, df = 1, p-value = 1.731e-12

\$`29`  
\$`29`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model  
F-Test = 2.3883, df1 = 10, df2 = 2130, p-value = 0.008165

\$`29`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model  
Chi-squared = 30.79, df = 1, p-value = 2.875e-08

\$`3`

\$`3`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 1.5633, df1 = 10, df2 = 2130, p-value = 0.1115

\$`3`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 90.209, df = 1, p-value < 2.2e-16

\$`30`

\$`30`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 3.298, df1 = 10, df2 = 2130, p-value = 0.0002956

\$`30`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 39.516, df = 1, p-value = 3.254e-10

\$`31`

\$`31`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 2.8515, df1 = 10, df2 = 2130, p-value = 0.001569

\$`31`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 62.183, df = 1, p-value = 3.109e-15

\$`32`

\$`32`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 2.3554, df1 = 10, df2 = 2130, p-value = 0.009142

\$`32`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 8.777, df = 1, p-value = 0.003051

\$`33`

\$`33`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 2.3574, df1 = 10, df2 = 2130, p-value = 0.009081

\$`33`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 12.298, df = 1, p-value = 0.0004534

\$`34`  
\$`34`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model  
F-Test = 1.5098, df1 = 10, df2 = 2130, p-value = 0.1294

\$`34`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model  
Chi-squared = 148.63, df = 1, p-value < 2.2e-16

\$`35`  
\$`35`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model  
F-Test = 4.5681, df1 = 10, df2 = 2130, p-value = 1.933e-06

\$`35`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model  
Chi-squared = 10.716, df = 1, p-value = 0.001062

\$`36`  
\$`36`\$Granger

Granger causality H0: x do not Granger-cause y

```
data: VAR object var_model
F-Test = 3.4864, df1 = 10, df2 = 2130, p-value = 0.0001434
```

```
$`36`$Instant
```

```
H0: No instantaneous causality between: x and y
```

```
data: VAR object var_model
Chi-squared = 53.387, df = 1, p-value = 2.74e-13
```

```
$`37`
$`37`$Granger
```

```
Granger causality H0: x do not Granger-cause y
```

```
data: VAR object var_model
F-Test = 4.0515, df1 = 10, df2 = 2130, p-value = 1.556e-05
```

```
$`37`$Instant
```

```
H0: No instantaneous causality between: x and y
```

```
data: VAR object var_model
Chi-squared = 7.7504, df = 1, p-value = 0.00537
```

```
$`38`
$`38`$Granger
```

```
Granger causality H0: x do not Granger-cause y
```

```
data: VAR object var_model
F-Test = 5.1319, df1 = 10, df2 = 2130, p-value = 1.896e-07
```

```
$`38`$Instant
```



H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 40.785, df = 1, p-value = 1.699e-10

\$`39`

\$`39`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 1.7147, df1 = 10, df2 = 2130, p-value = 0.07195

\$`39`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 0.045353, df = 1, p-value = 0.8314

\$`4`

\$`4`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 2.8366, df1 = 10, df2 = 2130, p-value = 0.001657

\$`4`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 3.0385, df = 1, p-value = 0.08131

\$`40`

\$`40`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 5.3177, df1 = 10, df2 = 2130, p-value = 8.751e-08

\$`40`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 16.144, df = 1, p-value = 5.869e-05

\$`41`

\$`41`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 1.9919, df1 = 10, df2 = 2130, p-value = 0.03057

\$`41`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 6.9095, df = 1, p-value = 0.008574

\$`42`

\$`42`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 3.0094, df1 = 10, df2 = 2130, p-value = 0.0008765

\$`42`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 48.365, df = 1, p-value = 3.539e-12

\$`5`

\$`5`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 2.4118, df1 = 10, df2 = 2130, p-value = 0.007527

\$`5`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 2.1408, df = 1, p-value = 0.1434

\$`6`

\$`6`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 2.6371, df1 = 10, df2 = 2130, p-value = 0.003408

\$`6`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 10.58, df = 1, p-value = 0.001143

\$`7`

\$`7`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 2.3975, df1 = 10, df2 = 2130, p-value = 0.007909

\$`7`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 17.798, df = 1, p-value = 2.457e-05

\$`8`

\$`8`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 1.6118, df1 = 10, df2 = 2130, p-value = 0.09715

\$`8`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 0.037436, df = 1, p-value = 0.8466

\$`9`

\$`9`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model  
F-Test = 2.1276, df1 = 10, df2 = 2130, p-value = 0.01967

\$`9`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model  
Chi-squared = 1.5116, df = 1, p-value = 0.2189

```
granger_test(df.ts.series2, y_col = "twd", x_col= "dp", group = "tree.id")
```

\$`1`

\$`1`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model  
F-Test = 5.1158, df1 = 10, df2 = 2130, p-value = 2.028e-07

\$`1`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model  
Chi-squared = 1.8342, df = 1, p-value = 0.1756

\$`10`

\$`10`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model  
F-Test = 3.2559, df1 = 10, df2 = 2130, p-value = 0.0003469

\$`10`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 21.65, df = 1, p-value = 3.272e-06

\$`11`

\$`11`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 6.388, df1 = 10, df2 = 2130, p-value = 9.515e-10

\$`11`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 75.179, df = 1, p-value < 2.2e-16

\$`12`

\$`12`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 8.5447, df1 = 10, df2 = 2130, p-value = 8.46e-14

\$`12`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 74.202, df = 1, p-value < 2.2e-16

\$`13`

\$`13`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 0.90368, df1 = 10, df2 = 2130, p-value = 0.5288

\$`13`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 1.2288, df = 1, p-value = 0.2676

\$`14`

\$`14`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 5.9657, df1 = 10, df2 = 2130, p-value = 5.73e-09

\$`14`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 73.104, df = 1, p-value < 2.2e-16

\$`15`

\$`15`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model  
F-Test = 4.3255, df1 = 10, df2 = 2130, p-value = 5.178e-06

\$`15`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model  
Chi-squared = 79.618, df = 1, p-value < 2.2e-16

\$`16`

\$`16`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model  
F-Test = 4.5113, df1 = 10, df2 = 2130, p-value = 2.437e-06

\$`16`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model  
Chi-squared = 38.857, df = 1, p-value = 4.56e-10

\$`17`

\$`17`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model  
F-Test = 5.7456, df1 = 10, df2 = 2130, p-value = 1.453e-08

\$`17`\$Instant

H0: No instantaneous causality between: x and y



data: VAR object var\_model  
Chi-squared = 27.535, df = 1, p-value = 1.543e-07

\$`18`  
\$`18`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model  
F-Test = 1.9354, df1 = 10, df2 = 2130, p-value = 0.03658

\$`18`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model  
Chi-squared = 0.043453, df = 1, p-value = 0.8349

\$`19`  
\$`19`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model  
F-Test = 7.0392, df1 = 10, df2 = 2130, p-value = 5.83e-11

\$`19`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model  
Chi-squared = 37.016, df = 1, p-value = 1.172e-09

\$`2`

\$`2`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 3.8633, df1 = 10, df2 = 2130, p-value = 3.286e-05

\$`2`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 4.3433, df = 1, p-value = 0.03715

\$`20`

\$`20`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 1.8019, df1 = 10, df2 = 2130, p-value = 0.05535

\$`20`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 0.000311, df = 1, p-value = 0.9859

\$`21`

\$`21`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 0.96087, df1 = 10, df2 = 2130, p-value = 0.4758

\$`21`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 14.282, df = 1, p-value = 0.0001574

\$`22`

\$`22`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 2.362, df1 = 10, df2 = 2130, p-value = 0.008938

\$`22`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 2.9681, df = 1, p-value = 0.08492

\$`23`

\$`23`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 2.84, df1 = 10, df2 = 2130, p-value = 0.001637

\$`23`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 16.168, df = 1, p-value = 5.797e-05

\$`24`

\$`24`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 4.5178, df1 = 10, df2 = 2130, p-value = 2.374e-06

\$`24`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 2.594, df = 1, p-value = 0.1073

\$`25`

\$`25`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 3.4608, df1 = 10, df2 = 2130, p-value = 0.0001584

\$`25`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 0.89816, df = 1, p-value = 0.3433

\$`26`

\$`26`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model  
F-Test = 4.472, df1 = 10, df2 = 2130, p-value = 2.859e-06

\$`26`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model  
Chi-squared = 0.23764, df = 1, p-value = 0.6259

\$`27`  
\$`27`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model  
F-Test = 3.8749, df1 = 10, df2 = 2130, p-value = 3.14e-05

\$`27`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model  
Chi-squared = 61.652, df = 1, p-value = 4.108e-15

\$`28`  
\$`28`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model  
F-Test = 6.6372, df1 = 10, df2 = 2130, p-value = 3.279e-10

\$`28`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 50.514, df = 1, p-value = 1.183e-12

\$`29`

\$`29`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 4.1582, df1 = 10, df2 = 2130, p-value = 1.015e-05

\$`29`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 30.798, df = 1, p-value = 2.864e-08

\$`3`

\$`3`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 5.045, df1 = 10, df2 = 2130, p-value = 2.721e-07

\$`3`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 0.53521, df = 1, p-value = 0.4644

\$`30`

\$`30`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 5.9784, df1 = 10, df2 = 2130, p-value = 5.432e-09

\$`30`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 42.569, df = 1, p-value = 6.824e-11

\$`31`

\$`31`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 5.3936, df1 = 10, df2 = 2130, p-value = 6.372e-08

\$`31`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 68.994, df = 1, p-value < 2.2e-16

\$`32`

\$`32`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 2.1541, df1 = 10, df2 = 2130, p-value = 0.01803

\$`32`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 16.272, df = 1, p-value = 5.487e-05

\$`33`

\$`33`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 2.5757, df1 = 10, df2 = 2130, p-value = 0.004239

\$`33`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 8.6508, df = 1, p-value = 0.003269

\$`34`

\$`34`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 4.4213, df1 = 10, df2 = 2130, p-value = 3.513e-06

\$`34`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model



Chi-squared = 0.0082372, df = 1, p-value = 0.9277

\$`35`

\$`35`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 3.6917, df1 = 10, df2 = 2130, p-value = 6.455e-05

\$`35`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 4.1384, df = 1, p-value = 0.04192

\$`36`

\$`36`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 3.8785, df1 = 10, df2 = 2130, p-value = 3.095e-05

\$`36`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 7.5323, df = 1, p-value = 0.00606

\$`37`

\$`37`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 5.1417, df1 = 10, df2 = 2130, p-value = 1.821e-07

\$`37`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 39.771, df = 1, p-value = 2.856e-10

\$`38`

\$`38`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 1.9489, df1 = 10, df2 = 2130, p-value = 0.03505

\$`38`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 0.33091, df = 1, p-value = 0.5651

\$`39`

\$`39`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 2.6636, df1 = 10, df2 = 2130, p-value = 0.0031

\$`39`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 34.707, df = 1, p-value = 3.832e-09

\$`4`

\$`4`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 2.9993, df1 = 10, df2 = 2130, p-value = 0.0009099

\$`4`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 13.187, df = 1, p-value = 0.0002819

\$`40`

\$`40`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 2.8864, df1 = 10, df2 = 2130, p-value = 0.001381

\$`40`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 0.00025877, df = 1, p-value = 0.9872

\$`41`

\$`41`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 3.792, df1 = 10, df2 = 2130, p-value = 4.353e-05

\$`41`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 1.0393, df = 1, p-value = 0.308

\$`42`

\$`42`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 3.8503, df1 = 10, df2 = 2130, p-value = 3.46e-05

\$`42`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 0.038362, df = 1, p-value = 0.8447

\$`5`

\$`5`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 3.0093, df1 = 10, df2 = 2130, p-value = 0.0008769

\$`5`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 14.578, df = 1, p-value = 0.0001344

\$`6`

\$`6`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 2.2431, df1 = 10, df2 = 2130, p-value = 0.01339

\$`6`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 21.758, df = 1, p-value = 3.093e-06

\$`7`

\$`7`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 1.2934, df1 = 10, df2 = 2130, p-value = 0.2282

\$`7`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model  
Chi-squared = 0.4058, df = 1, p-value = 0.5241

\$`8`  
\$`8`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model  
F-Test = 2.6505, df1 = 10, df2 = 2130, p-value = 0.003248

\$`8`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model  
Chi-squared = 2.1941, df = 1, p-value = 0.1385

\$`9`  
\$`9`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model  
F-Test = 4.0296, df1 = 10, df2 = 2130, p-value = 1.698e-05

\$`9`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model  
Chi-squared = 15.262, df = 1, p-value = 9.356e-05

## Granger test by species

```
granger_test(df.ts.series2, y_col= "twd", x_col= "at", group = "species")
```

```
$`Carpinus betulus`  
$`Carpinus betulus`$Granger
```

Granger causality H0: x do not Granger-cause y

```
data: VAR object var_model  
F-Test = 7.3404, df1 = 10, df2 = 8706, p-value = 1.099e-11
```

```
$`Carpinus betulus`$Instant
```

H0: No instantaneous causality between: x and y

```
data: VAR object var_model  
Chi-squared = 24.933, df = 1, p-value = 5.937e-07
```

```
$`Corylus avellana`  
$`Corylus avellana`$Granger
```

Granger causality H0: x do not Granger-cause y

```
data: VAR object var_model  
F-Test = 2.68, df1 = 10, df2 = 4322, p-value = 0.002861
```

```
$`Corylus avellana`$Instant
```

H0: No instantaneous causality between: x and y

```
data: VAR object var_model  
Chi-squared = 0.10641, df = 1, p-value = 0.7443
```

```
$`Fagus sylvatica`
```

\$`Fagus sylvatica`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 6.9531, df1 = 10, df2 = 21858, p-value = 5.704e-11

\$`Fagus sylvatica`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 311.05, df = 1, p-value < 2.2e-16

\$`Picea abies`

\$`Picea abies`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 13.061, df1 = 10, df2 = 32818, p-value < 2.2e-16

\$`Picea abies`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 48.338, df = 1, p-value = 3.588e-12

\$`Pinus sylvestris`

\$`Pinus sylvestris`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 8.3564, df1 = 10, df2 = 10898, p-value = 1.141e-13



\$`Pinus sylvestris`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 272.9, df = 1, p-value < 2.2e-16

\$`Pseudotsuga menziesii`

\$`Pseudotsuga menziesii`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 1.9786, df1 = 10, df2 = 4322, p-value = 0.03161

\$`Pseudotsuga menziesii`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 15.741, df = 1, p-value = 7.264e-05

\$`Quercus petraea`

\$`Quercus petraea`\$Granger

Granger causality H0: x do not Granger-cause y

data: VAR object var\_model

F-Test = 3.9166, df1 = 10, df2 = 8706, p-value = 2.443e-05

\$`Quercus petraea`\$Instant

H0: No instantaneous causality between: x and y

data: VAR object var\_model

Chi-squared = 184.65, df = 1, p-value < 2.2e-16

## Linear regression

```
plot_coeffs <- function(my_coef_df){  
  ggplot(my_coef_df, aes(x = reorder(term, estimate), y = estimate)) +  
    geom_bar(stat = "identity", fill = "steelblue") +  
    geom_errorbar(aes(ymin = estimate - std.error, ymax = estimate + std.error), width = 0.2,  
    coord_flip() + # flip axes for readability  
    labs(  
      title = "Linear Model Coefficients",  
      x = "Predictor",  
      y = "Estimate"  
    ) +  
    theme_minimal()  
}
```

```
lm_fit <- lm(  
  formula = twd ~ cum.time.days + at + dp + pr + ws + sr + lr,  
  data = df.ts.series2  
)  
  
summary(lm_fit)
```

Call:

```
lm(formula = twd ~ cum.time.days + at + dp + pr + ws + sr + lr,  
    data = df.ts.series2)
```

Residuals:

|  | Min     | 1Q      | Median  | 3Q     | Max    |
|--|---------|---------|---------|--------|--------|
|  | -2.1275 | -0.5348 | -0.2398 | 0.2686 | 6.5412 |

Coefficients:

|               | Estimate   | Std. Error | t value | Pr(> t )     |
|---------------|------------|------------|---------|--------------|
| (Intercept)   | -7.059e-02 | 6.781e-02  | -1.041  | 0.298        |
| cum.time.days | 3.474e-04  | 1.364e-05  | 25.477  | < 2e-16 ***  |
| at            | 7.014e-02  | 1.826e-03  | 38.412  | < 2e-16 ***  |
| dp            | -1.184e-01 | 2.142e-03  | -55.269 | < 2e-16 ***  |
| pr            | -1.551e-01 | 2.436e-02  | -6.365  | 1.97e-10 *** |
| ws            | -1.024e-01 | 3.145e-03  | -32.578 | < 2e-16 ***  |
| sr            | -3.721e-04 | 7.651e-05  | -4.863  | 1.16e-06 *** |
| lr            | 2.454e-03  | 2.354e-04  | 10.422  | < 2e-16 ***  |

---

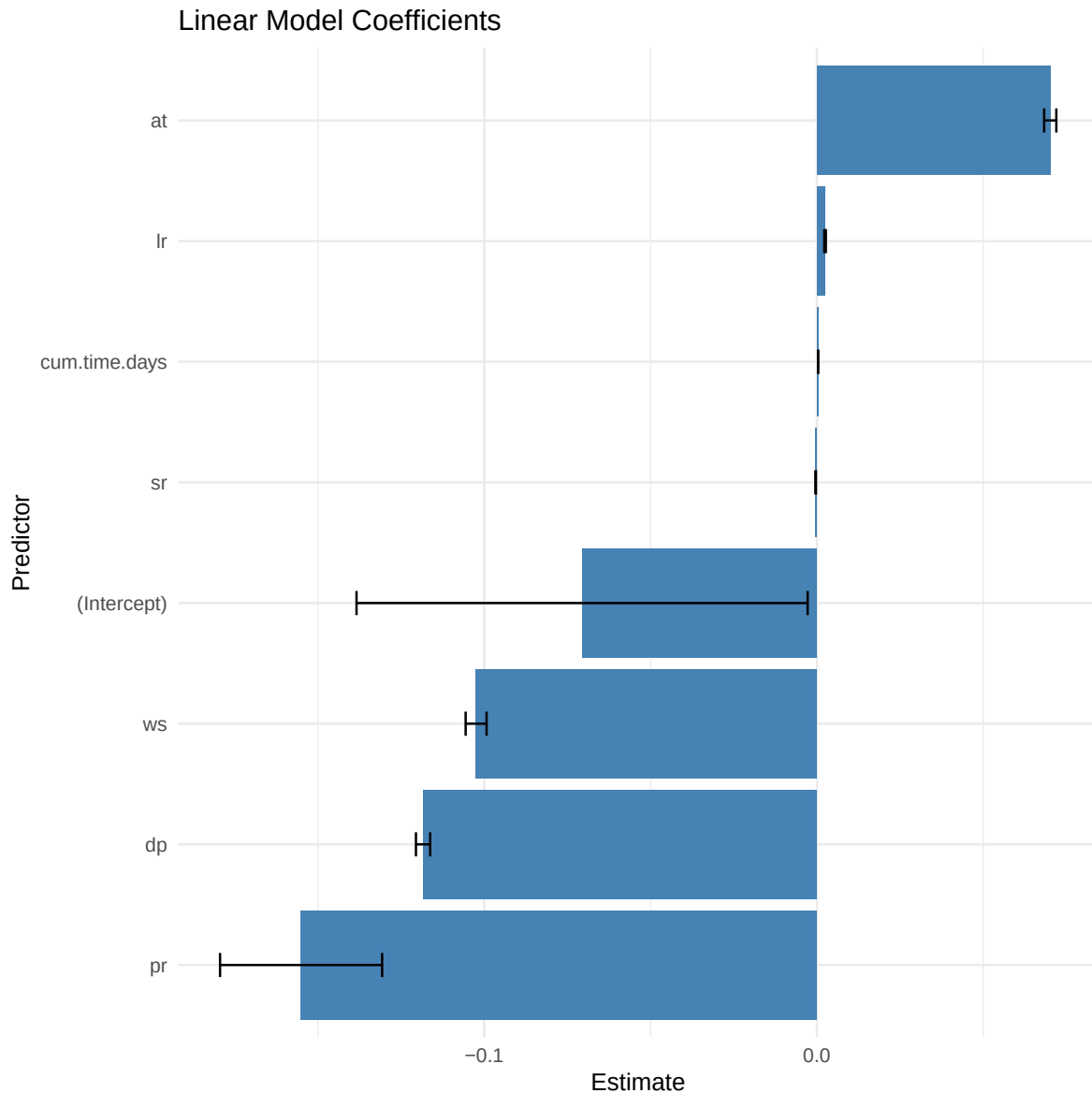
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9105 on 46024 degrees of freedom

Multiple R-squared: 0.1216, Adjusted R-squared: 0.1215

F-statistic: 910.5 on 7 and 46024 DF, p-value: < 2.2e-16

```
coef_df <- tidy(lm_fit) # gets estimate, std.error, etc.  
plot_coefs(coef_df)
```



```
lm_fit <- lm(
  formula = twd ~ cum.time.days + at + dp + pr + ws + sr + lr,
  data = df.ts.series2,
  subset = species %in% c("Picea abies")
)

summary(lm_fit)
```

Call:

```
lm(formula = twd ~ cum.time.days + at + dp + pr + ws + sr + lr,
    data = df.ts.series2, subset = species %in% c("Picea abies"))
```

Residuals:

| Min     | 1Q      | Median  | 3Q     | Max    |
|---------|---------|---------|--------|--------|
| -2.3254 | -0.6140 | -0.2014 | 0.3965 | 5.6093 |

Coefficients:

|               | Estimate   | Std. Error | t value | Pr(> t )     |
|---------------|------------|------------|---------|--------------|
| (Intercept)   | 9.549e-02  | 1.218e-01  | 0.784   | 0.433121     |
| cum.time.days | 2.939e-04  | 2.330e-05  | 12.615  | < 2e-16 ***  |
| at            | 3.801e-02  | 2.937e-03  | 12.942  | < 2e-16 ***  |
| dp            | -9.928e-02 | 3.999e-03  | -24.826 | < 2e-16 ***  |
| pr            | -1.492e-01 | 4.416e-02  | -3.378  | 0.000732 *** |
| ws            | -7.186e-03 | 6.952e-03  | -1.034  | 0.301363     |
| sr            | 8.313e-05  | 1.170e-04  | 0.711   | 0.477347     |
| lr            | 2.132e-03  | 4.145e-04  | 5.144   | 2.72e-07 *** |

---

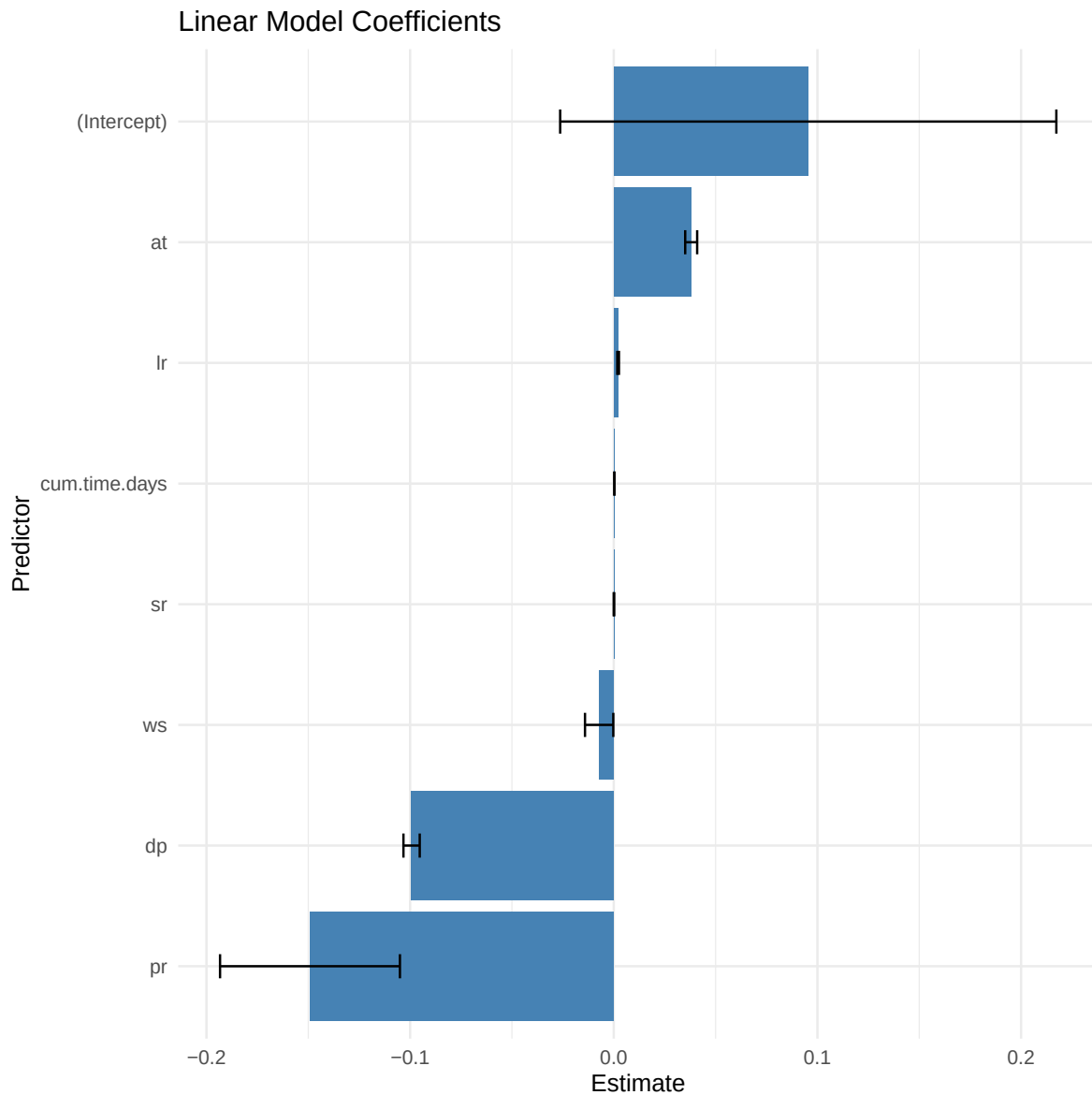
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9286 on 16432 degrees of freedom

Multiple R-squared: 0.1282, Adjusted R-squared: 0.1278

F-statistic: 345.2 on 7 and 16432 DF, p-value: < 2.2e-16

```
coef_df <- tidy(lm_fit) # gets estimate, std.error, etc.
plot_coefs(coef_df)
```



```
lm_fit <- lm(  
  formula = twd ~ cum.time.days + at + dp + pr + ws + sr + lr,  
  data = df.ts.series2,  
  subset = species %in% c("Fagus sylvatica")  
)  
  
summary(lm_fit)
```

```
Call:
lm(formula = twd ~ cum.time.days + at + dp + pr + ws + sr + lr,
    data = df.ts.series2, subset = species %in% c("Fagus sylvatica"))
```

Residuals:

| Min     | 1Q      | Median  | 3Q     | Max    |
|---------|---------|---------|--------|--------|
| -1.5274 | -0.3262 | -0.1401 | 0.1694 | 4.9716 |

Coefficients:

|               | Estimate   | Std. Error | t value | Pr(> t )     |
|---------------|------------|------------|---------|--------------|
| (Intercept)   | 4.064e-01  | 9.252e-02  | 4.392   | 1.13e-05 *** |
| cum.time.days | 2.595e-04  | 1.823e-05  | 14.234  | < 2e-16 ***  |
| at            | 9.608e-02  | 2.805e-03  | 34.256  | < 2e-16 ***  |
| dp            | -1.051e-01 | 3.134e-03  | -33.550 | < 2e-16 ***  |
| pr            | -1.054e-01 | 2.865e-02  | -3.678  | 0.000236 *** |
| ws            | -3.170e-02 | 4.453e-03  | -7.119  | 1.16e-12 *** |
| sr            | -1.215e-03 | 1.195e-04  | -10.170 | < 2e-16 ***  |
| lr            | -8.737e-04 | 3.253e-04  | -2.686  | 0.007242 **  |

---

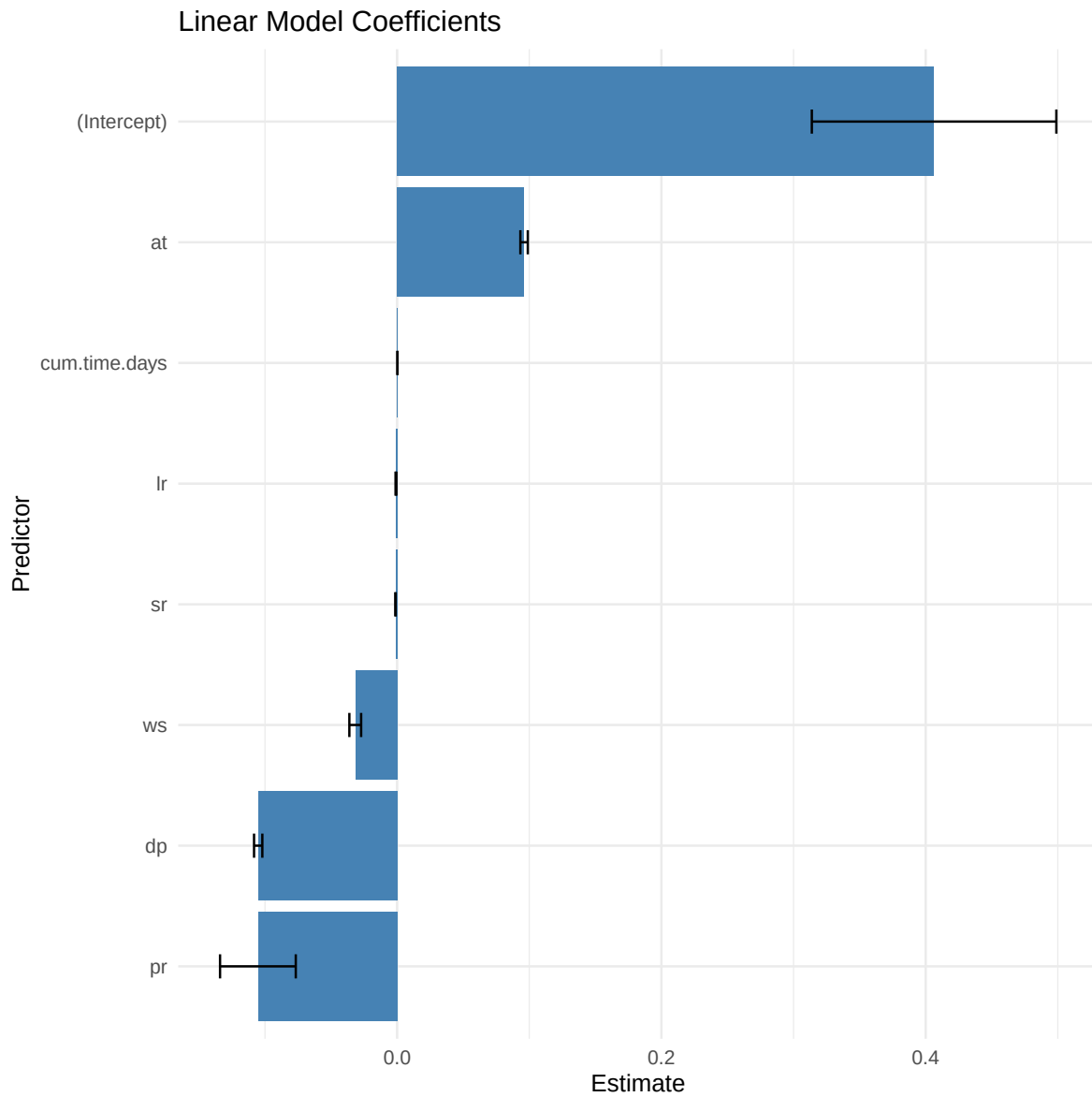
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5929 on 10952 degrees of freedom

Multiple R-squared: 0.1739, Adjusted R-squared: 0.1734

F-statistic: 329.3 on 7 and 10952 DF, p-value: < 2.2e-16

```
coef_df <- tidy(lm_fit) # gets estimate, std.error, etc.
plot_coefs(coef_df)
```



## Rollins SD

Confirm that timeseries are not stationary

```
library(dplyr)
library(purrr)
```

Attaching package: 'purrr'

The following object is masked from 'package:car':

some

```
library(ggplot2)
library(broom)
library(tidyr)
library(ggplot2)

# Nest data by tree.id
nested <- df.ts.series2 %>%
  group_by(tree.id) %>%
  nest()

# Fit model and extract residuals
residuals_by_tree <- nested %>%
  mutate(
    model = map(data, ~ lm(twd ~ cum.time.days, data = .x)),
    augmented = map(model, augment)
  ) %>%
  unnest(augmented)

residuals_by_tree_p1 <- residuals_by_tree %>%
  filter(tree.id < 22)
residuals_by_tree_p2 <- residuals_by_tree %>%
  filter(tree.id > 21)

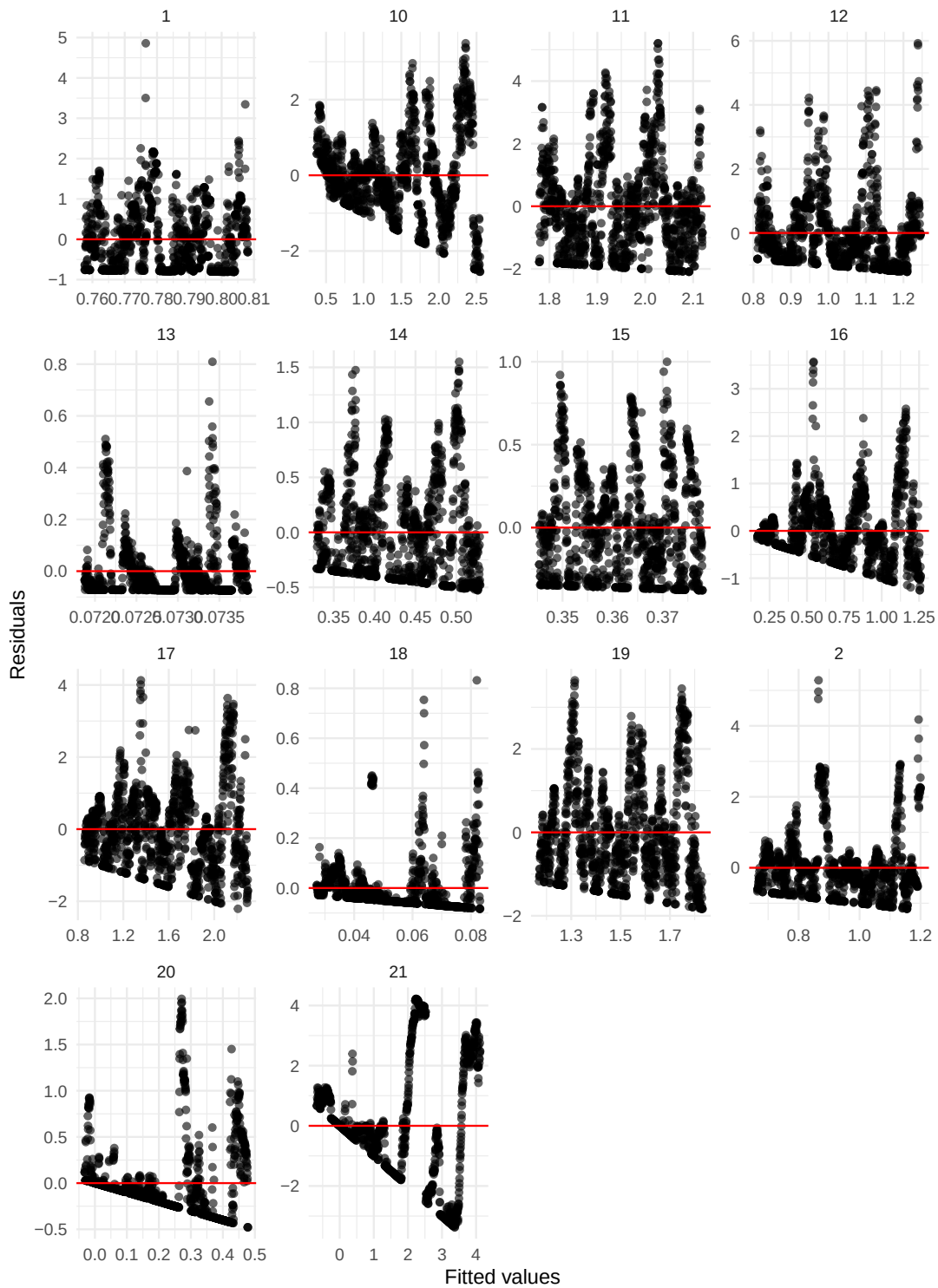
plot_residuals <- function(df){
  # Plot residuals vs fitted for each tree
  print(ggplot(df, aes(.fitted, .resid)) +
    geom_point(alpha = 0.6) +
    geom_hline(yintercept = 0, color = "red") +
    facet_wrap(~ tree.id, scales = "free", ncol = 4) +
    labs(x = "Fitted values", y = "Residuals",
         title = "Residuals vs Fitted Values for Each Tree ID") +
    theme_minimal())

  print(ggplot(df, aes(x = cum.time.days, y = .resid)) +
    geom_point(alpha = 0.6) +
```

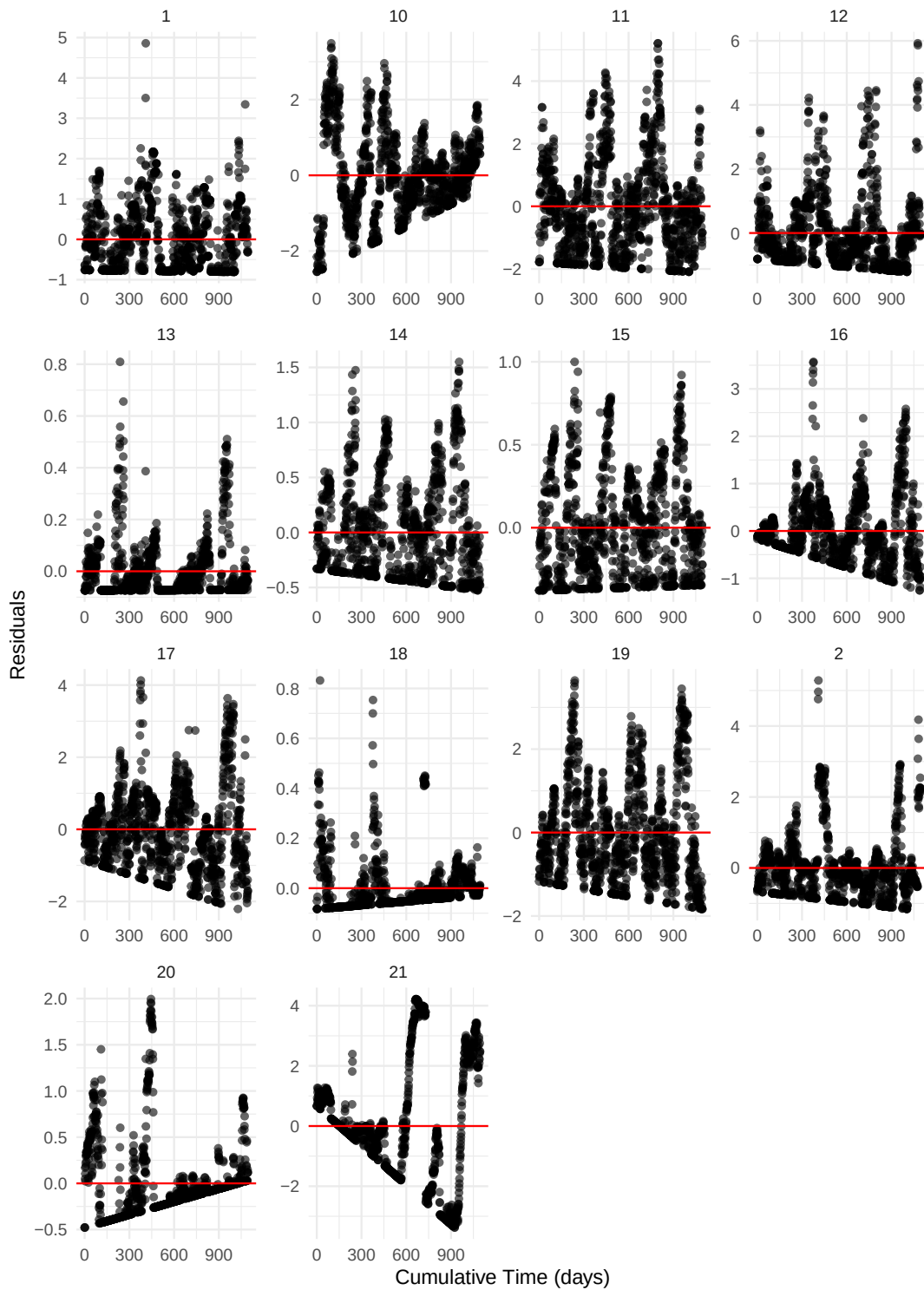


```
geom_hline(yintercept = 0, color = "red") +  
facet_wrap(~ tree.id, scales = "free", ncol = 4) +  
labs(x = "Cumulative Time (days)", y = "Residuals",  
      title = "Residuals vs. Cumulative Time by Tree ID") +  
theme_minimal()  
}  
  
plot_residuals(residuals_by_tree_p1)
```

Residuals vs Fitted Values for Each Tree ID

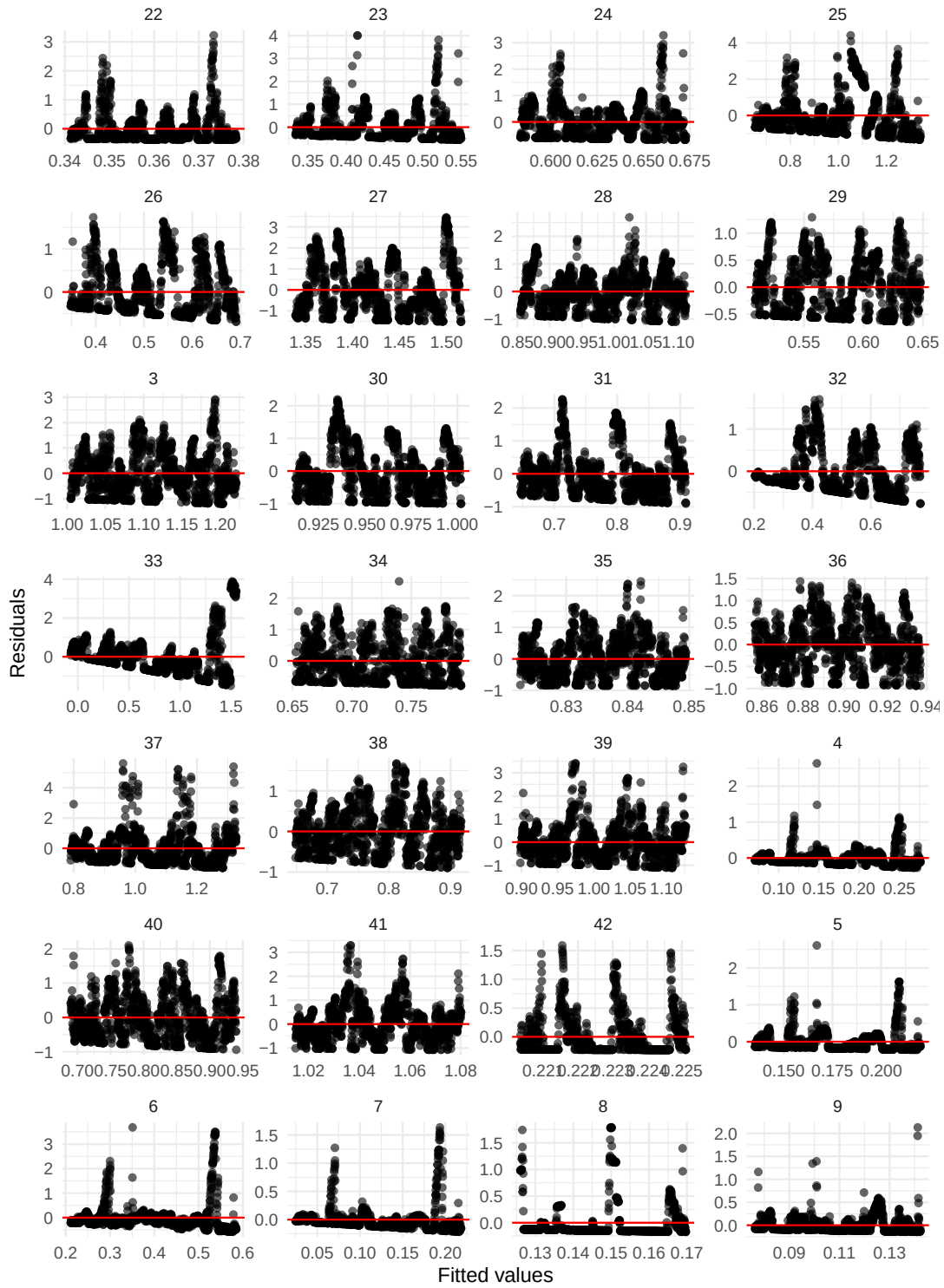


Residuals vs. Cumulative Time by Tree ID

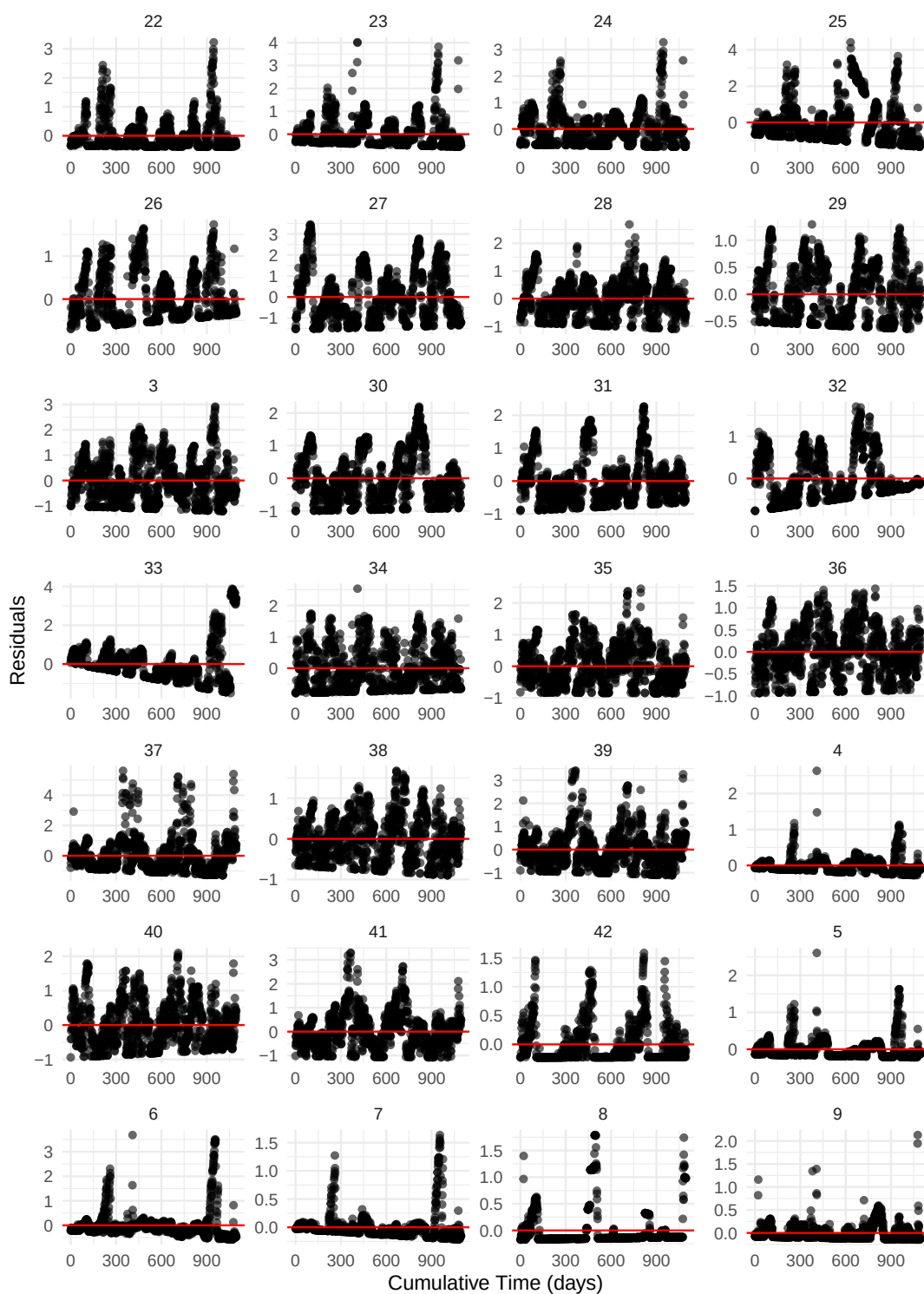


```
plot_residuals(residuals_by_tree_p2)
```

Residuals vs Fitted Values for Each Tree ID



Residuals vs. Cumulative Time by Tree ID



```

# Step 1: Fit a linear model for each tree
coeffs_by_tree <- df.ts.series2 %>%
  group_by(tree.id) %>%
  nest() %>%
  mutate(
    model = map(data, ~ lm(twd ~ cum.time.days, data = .x)),
    coef = map(model, tidy)
  ) %>%
  unnest(coef)

# Step 2: Plot coefficients
ggplot(coeffs_by_tree, aes(x = tree.id, y = estimate, fill = term)) +
  geom_bar(stat = "identity", position = position_dodge()) +
  labs(x = "Tree ID", y = "Coefficient Estimate", fill = "Term",
       title = "Linear Model Coefficients by Tree ID") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 90))

```

